

Quantum Cohomology

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Abstract

Quantum cohomology arises as a deformation of the cup product in the classical cohomology ring; its study was originally motivated by developments in theoretical physics. In this thesis, we present a construction of quantum cohomology for homogeneous varieties. To build quantum cohomology, we first review the moduli spaces of curves and maps, alongside their respective compactifications due to Deligne-Mumford and Kontsevich. We then explore the (small) quantum cohomology of the Grassmannian. In this setting, Schubert classes allow us to compute the quantum cohomology ring explicitly. We review the classical Pieri and Giambelli formulas, which provide explicit combinatorial formulas for computing products of Schubert classes, and conclude by discussing their quantum analogues.

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1 Introduction

The numbers now known as Gromov-Witten invariants originated in physics, where they were used to compute quantum corrections to Yukawa couplings in string theory via the A-model. These corrections arose from the various ways a string could wrap around a target Calabi-Yau manifold (serving as the compactified dimensions of spacetime). Physicists had an alternative method for computing these couplings by using the so-called B-model on a “mirror” geometry. After passing through the “mirror map,” the B-model approach yielded a power series whose coefficients encoded the Gromov-Witten invariants. Physicists P. Candelas, X. C. de la Ossa, P. S. Green, and L. Parkes [Can+91] constructed the “mirror” for the case where the Calabi-Yau is a quintic threefold. By computing the first coefficients from the power series coming from the B-model on the mirror, they gave predictions for a long-standing question in enumerative geometry: what is the number of rational curves on a general quintic threefold? They correctly predicted the well-known number of lines in a quintic threefold, 2875, and their predictions for higher degrees were subsequently verified. At the time, computing these numbers felt completely out of reach of mathematics. In an effort to formalize and understand these results from physics, Kontsevich introduced the moduli space of stable maps, the first key step in placing Gromov-Witten

invariants on a firm mathematical footing [Kon95]. In the context of homogeneous spaces, the moduli space of stable maps provides a complete foundation for the theory of genus zero Gromov-Witten invariants.

The idea of quantum cohomology is to then use genus zero Gromov-Witten invariants to provide a deformation of the classical cup product in cohomology. The way in which this is done is through a “potential function” motivated by physics. In this thesis, we provide a brief introduction to the moduli of stable maps, Gromov-Witten invariants, and quantum cohomology. We will then study a variation called small quantum cohomology. This variation is easier to study and leads to rich combinatorics in the case of the Grassmannian. The rest of the thesis will focus on exploring the generalizations of the Pieri and Giambelli formulas from classical Schubert calculus to the (small) quantum cohomology of the Grassmannian. All our schemes will be over $\mathbb{C}$. The needed background on moduli theory and intersection theory is reviewed in the appendices.

2 Grassmannians

Homogeneous spaces form a nice class of varieties over which Gromov-Witten invariants can be defined in a simple manner. In this section, we review the definition of homogeneous spaces, with the Grassmannian as our example. Further, as a precursor to studying the quantum cohomology of the Grassmannian we will discuss its classical cohomology. We will see that the Grassmannian can be stratified using *Schubert cells*, the closures of which give us *Schubert varieties* defining cycles on $A_*(\mathrm{Gr}(k, n))$. The Poincare duals of these Schubert cycles form a $\mathbb{Z}$ -basis for $A^*(\mathrm{Gr}(k, n))$. We will review the classical Pieri and Giambelli formulas, which provide a combinatorial method for computing the classical product of Schubert classes. In section 5 we use the Schubert classes to describe the quantum cohomology of the Grassmannian. Our discussion of homogeneous spaces is inspired by Michel Brion’s notes on actions of algebraic groups [Bri10], and Victoria Hoskins’ notes on geometric invariant theory [Hos23]. Our discussion of Schubert Calculus is inspired by Chapter 3 of [EH16].

2.1 Algebraic Groups

First let us start with the definition of algebraic groups over $\mathrm{Spec}(\mathbb{C})$.

Definition 2.1. An *algebraic group* over $\mathbb{C}$ is a variety G over $\mathbb{C}$ with morphisms $e : \mathrm{Spec}(\mathbb{C}) \rightarrow G$, multiplication $m : G \times G \rightarrow G$, and inverse $i : G \rightarrow G$, such that m and i fit into commuting diagrams that justify calling m “multiplication”,

and i “inverse”. See [Hos23] for details.

If we only consider the closed points of G as a set, denoted by $G(\mathbb{C})$, then m , i , and e , endow $G(\mathbb{C})$ with the structure of a group.

Example 2.2. Let G be the general linear group $\mathrm{GL}_n(\mathbb{C})$. We can write down G as an algebraic variety by using the fact that it is the complement of the vanishing locus of the determinant on $\mathrm{Mat}_{n \times n}(\mathbb{C}) \simeq \mathbb{C}^{n^2}$. In other words, we have $G = \mathrm{Spec}(\mathbb{C}[x_{i,j} : 1 \leq i, j \leq n]_{\det(x_{ij})})$.

Just as ordinary groups can act on sets, algebraic groups can act on varieties.

Definition 2.3. A G -variety is a variety X equipped with an action of the algebraic group G . That is, there is a morphism

$$\alpha : G \times X \rightarrow X, \quad (g, x) \rightarrow g \cdot x,$$

satisfying the expected commuting diagrams for an action. See [Hos23] for details.

The action of an algebraic group G on a variety X induces an action of the group $G(\mathbb{C})$ on the set of closed points of X , denoted as $X(\mathbb{C})$. We say that the action of G is transitive if the induced action of $G(\mathbb{C})$ on $X(\mathbb{C})$ is transitive.

Definition 2.4. A variety X is *homogeneous* if it is G -variety where the action of G is transitive.

Remark 2.5. We can provide an explicit description of homogeneous varieties as quotients of algebraic groups. From group theory, we know that all transitive G -sets are isomorphic to the action of G on the cosets G/H for some subgroup $H \subset G$. The geometric analogue of this classification tells us that if X is a homogeneous variety, then $X \simeq G/H$ for some *closed* subgroup $H \subset G$. The key here is that if $H \subset G$ is closed then G/H can be given a natural structure as G -variety. See Theorem 1.16 in [Bri10].

Our main example of a homogeneous variety will be the Grassmannian. Let $G = \mathrm{GL}_n(\mathbb{C})$, and $\mathrm{Gr}(k, n)$ be the Grassmannian of k -dimensional subspaces of $\mathbb{C}^n$. There is a natural action of $\mathrm{GL}_n(\mathbb{C})$ on $\mathrm{Gr}(k, n)$ given by taking a k -dimensional subspace $V \subset \mathbb{C}^n$ and acting on it by an automorphism $g : \mathbb{C}^n \rightarrow \mathbb{C}^n$. In fact, any k -dimensional subspace $W \subset \mathbb{C}^n$ can be obtained from a fixed $V \subset \mathbb{C}^n$ through an automorphism $g \in G$ (i.e G acts transitively on $\mathrm{Gr}(k, n)$). Therefore $\mathrm{Gr}(k, n)$ is a homogeneous variety.

Theorem 3.21 on the moduli of stable maps, and the construction of quantum cohomology for the Grassmannian in Chapter 4, both apply similarly to homogeneous varieties. Our focus on the Grassmannian is two-fold: first, for the sake of concreteness, and second, to study the simpler version of the quantum Pieri and Giambelli formulas that arise in this context. We proceed to study the Grassmannian.

The Grassmannian is in fact a *projective* variety. This can be seen through the Plücker embedding. Given a k -dimensional subspace $V \in \text{Gr}(k, n)$, let $v_1, \dots, v_k$ be a basis for V . Consider the map $V \mapsto v_1 \wedge v_2 \wedge \dots \wedge v_k$. Let $w_1, \dots, w_k$ be a different choice of basis for V . We have that

$$v_1 \wedge \dots \wedge v_k = \det(M) (w_1 \wedge \dots \wedge w_k),$$

where M is a change of basis matrix. Thus we get a well-defined map

$$\iota : \text{Gr}(k, n) \rightarrow \mathbb{P}(\bigwedge^k \mathbb{C}^n), \quad V \mapsto v_1 \wedge \dots \wedge v_k.$$

We call ι the Plücker embedding. In Chapter 3, we use the Plücker embedding to push forward a computation in $A^*(\text{Gr}(k, n))$ to $A^*(\mathbb{P}^N)$; see Example 3.22. We proceed to studying the cohomology of the Grassmannian.

2.2 Cohomology of the Grassmannian

In this section we let $X = \text{Gr}(k, n)$. Schubert varieties are an important class of subvarieties of the Grassmannian; their representatives in cohomology form a $\mathbb{Z}$ -basis for the cohomology ring. Schubert varieties are indexed by partitions $\lambda = (n - k \geq \lambda_1 \geq \dots \geq \lambda_k \geq 0)$, with $\lambda_i \in \mathbb{N}$. We can visualize a partition through its *Young diagram*. It is simplest to illustrate what a Young diagram is through examples. Let $\lambda = (3, 2, 1)$ and $\mu = (2, 1, 1)$. The corresponding Young diagrams are given by

$$\lambda = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \\ \hline \square & & \\ \hline \end{array} \qquad \mu = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \square & \\ \hline \end{array}.$$

Given a complete flag $F_\bullet$ and a partition $\lambda = (\lambda_1 \geq \dots \geq \lambda_k \geq 0)$ whose Young diagram fits inside an $k \times (n - k)$ rectangle (meaning $\lambda_1 \leq n - k$), we define the Schubert variety

$$(1) \quad \Omega_\lambda(F_\bullet) = \{V \in X : \dim(V \cap F_{(n-k)+i-\lambda_i}) \geq i \quad \forall 1 \leq i \leq k\}.$$

By Poincare duality each Schubert variety defines a *Schubert class* in $A^*(X)$. The Schubert class is independent of the choice of flag $F_\bullet$. For a Schubert variety $\Omega_\lambda(F_\bullet)$ we denote its corresponding Schubert class by $\Omega_\lambda \in A^{|\lambda|}(X)$; where

$$|\lambda| = \lambda_1 + \lambda_2 + \cdots + \lambda_k.$$

The Schubert classes Ω_λ form a $\mathbb{Z}$ -basis for $A^*(X)$. Moreover, we have a presentation for $A^*(X)$ as a ring

$$A^*(X) = \mathbb{Z}[\Omega_1, \dots, \Omega_{n-k}] / (Y_{k+1}, \dots, Y_n).$$

We use Ω_i for the Schubert class corresponding to the single row partition $\lambda = (i)$. Such Ω_i are called *special Schubert classes*. The relations are given by

$$Y_p = \det(\Omega_{1+j-i})_{1 \leq i, j \leq p} \quad \text{for } p = 1, \dots, n$$

For simplicity, the notation $\Omega_i = 0$ for $i < 0$, or $i > n - k$ is adopted. Pieri's formula is a combinatorial rule for computing the product of a Schubert class Ω_λ with a special Schubert class. Because special Schubert classes generate the ring $A^*(X)$, we may express every Ω_λ as the sum of products of special Schubert classes; this is the subject of the Giambelli Formula. Using the Pieri and Giambelli formulas we can compute the product of any two Schubert classes by expanding them out into sums of products of special Schubert classes.

Theorem 2.6 (Pieri's Formula). *Let Ω_p, Ω_μ be Schubert cycles. We have that*

$$\Omega_p \cdot \Omega_\mu = \sum_{\substack{\mu_i \leq \nu_i \leq \mu_{i-1} \\ |\nu| = p + |\mu|}} \Omega_\nu.$$

Giambelli's formula can be deduced as a Corollary of Pieri's formula.

Corollary 2.7 (Giambelli's Formula). *Let λ be a partition fitting inside a $k \times (n - k)$ rectangle. From Pieri's formula it follows that*

$$\Omega_\lambda = \det(\Omega_{\lambda_i + j - i})_{1 \leq i, j \leq k}.$$

Before moving on to quantum cohomology let us work out an explicit example on how these rules allow us to determine the product on $A^*(X)$.

Example 2.8. Let $n = 6, k = 3$. So we are considering Schubert classes in $\text{Gr}(3, 6)$. Fix the partitions $\lambda = (1, 1, 1), \mu = (2, 1, 0)$ with Young diagrams

$$\lambda = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \quad \mu = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array}.$$

Our goal is to compute $\Omega_\lambda \cdot \Omega_\mu$ using the Pieri and Giambelli Formula. First we use the Giambelli formula on Ω_μ to write it as a product of special Schubert classes

$$\Omega_{\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}} = \det \begin{bmatrix} \Omega_2 & \Omega_3 & \Omega_4 \\ \Omega_0 & \Omega_1 & \Omega_2 \\ \Omega_{-2} & \Omega_{-1} & \Omega_0 \end{bmatrix} = -\Omega_3 + \Omega_2 \cdot \Omega_1.$$

Here we use the fact that $\Omega_0 = 1$, and $\Omega_p = 0$ for $p < 0$ or $p > 3$. Now we must compute

$$(2) \quad \Omega_\lambda \cdot \Omega_\mu = -\Omega_\lambda \cdot \Omega_3 + (\Omega_\lambda \cdot \Omega_2) \cdot \Omega_1.$$

Let us compute each term separately, starting with $\Omega_\lambda \cdot \Omega_3$. Pieri's formula tells us to look at partitions ν fitting in a $k \times (n-k)$ rectangle that can be obtained from $\begin{array}{|c|} \hline \square \\ \hline \end{array}$ by adding 3 boxes with no two in the same column. Observe that there exists no such ν . It follows that

$$\Omega_\lambda \cdot \Omega_3 = 0.$$

Now we turn our attention to $\Omega_\lambda \cdot \Omega_2$. Pieri's formula tells us to look at partitions ν fitting in a $k \times (n-k)$ rectangle that can be obtained from $\begin{array}{|c|} \hline \square \\ \hline \end{array}$ by adding 2 boxes with no two in the same column. There exists one only such ν , namely $\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array}$; hence

$$\Omega_\lambda \cdot \Omega_2 = \Omega_{\begin{array}{|c|} \hline \square \\ \hline \end{array}} \cdot \Omega_{\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array}} = \Omega_{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \end{array}}.$$

Now we must compute $\Omega_{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \end{array}} \cdot \Omega_1$. Through a similar analysis using Pieri's rule we arrive at

$$\Omega_{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \end{array}} \cdot \Omega_1 = \Omega_{\begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & & & \\ \hline \end{array}}.$$

Substituting our expressions for $\Omega_\lambda \cdot \Omega_3$ and $\Omega_\lambda \cdot \Omega_2 \cdot \Omega_1$ back into (2) we conclude

$$\Omega_\lambda \cdot \Omega_\mu = \Omega_{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \square & & \\ \hline \end{array}} .$$

As a final note we remark that Schubert varieties arise as the closures of *Schubert cells*. Given a complete flag $F_\bullet$ and a partition λ whose Young diagram fits inside a $k \times (n - k)$ rectangle, we define the *Schubert cell*

$$(3) \quad \Omega_\lambda^\circ(F_\bullet) = \{V \in X \mid \dim(V \cap F_j) = i \text{ for } n - k + i - \lambda_i \leq j \leq n - k + i - \lambda_{i+1} \text{ for all } i\} .$$

The Grassmannian can be decomposed as a disjoint union across all Schubert cells indexed by partitions λ fitting inside a $k \times (n - k)$ rectangle. This is called the Schubert stratification of the Grassmannian. The Schubert stratification can be used to show the closure of the Schubert cells, the Schubert varieties, form a $\mathbb{Z}$ -basis for the cohomology ring of the Grassmannian.

Remark 2.9. In [EH16], Schubert cells $\Omega_\lambda^\circ(F_\bullet)$ are defined as the complement of a union of “smaller” Schubert varieties within $\Omega_\lambda(F_\bullet)$, see Section 4.1.2. Thus we should think of the boundary of a Schubert variety of consisting of smaller Schubert varieties. Lemma 4.5 in [EH16] then tells us that for general flag $F_\bullet, G_\bullet$, a general point $W \in \Omega_\lambda(F_\bullet) \cap \Omega_\mu(G_\bullet)$, lies in $W \in \Omega_\lambda(F_\bullet)^\circ \cap \Omega_\mu^\circ(G_\bullet)$.

Remark 2.10. In Appendix B, we discuss the Chow group $A_*(X)$ alongside its cohomology counterpart, operational Chow $A^*(X)$. In the case of homogeneous varieties, these are isomorphic to the topological homology and cohomology groups, $H_*(X)$ and $H^*(X)$. In fact, $A_i(X) \simeq H_{2i}(X)$ and $A^i(X) \simeq H^{2i}(X)$. Owing to this isomorphism, we will simply work with $A_i(X)$ and $A^i(X)$ and refer to them as homology and cohomology, respectively.

We now shift gears to discuss the moduli of curves and the moduli of stable maps. We return to the Grassmannian in Section 5.

3 Moduli of Curves and Stable Maps

In this section we will study the moduli space of n -pointed smooth rational projective curves $M_{0,n}$ and its Deligne-Mumford compactification $\overline{M}_{0,n}$. We then proceed to study the moduli of n -pointed maps $\mathbb{P}^1 \rightarrow X$ where the fundamental class $[\mathbb{P}^1]$ lands in a fixed $\beta \in A_1(X)$. The class β can be thought of as the degree of the map. We then study its compactification $\overline{M}_{0,n}(X, \beta)$ by Kontsevich. The

exposition in this thesis is inspired by [FP97] and [KV07]. In Appendix A we review the basic concepts of moduli theory used in this section.

3.1 Moduli of Curves

The first step in framing a moduli problem is to provide a definition of families. We begin with the definitions of families of n points in $\mathbb{P}^1$; we refer to these families of n -pointed curves.

Definition 3.1. Let S be a scheme. A *family of n -pointed curves over S* consists of

- A flat proper morphism $\pi : C \rightarrow S$, such that each geometric fiber $C_{\bar{s}}$ is isomorphic to $\mathbb{P}^1$.
- n disjoint sections $\sigma_i : S \rightarrow C$.

Definition 3.2. We say two families $C \rightarrow S$, $C' \rightarrow S$ are isomorphic if there exists an isomorphism $\varphi : C \rightarrow C'$ such that the following diagram commutes:

$$\begin{array}{ccc} C & \xrightarrow{\varphi} & C' \\ \searrow & & \swarrow \\ & S & \\ \swarrow & & \searrow \\ & S & \end{array}$$

σ_i σ'_i

We package these two definitions together in the definition of our moduli functor $\mathcal{M}_{0,n}$.

Definition 3.3. Let $\mathcal{M}_{0,n} : \text{Sch}^{op} \rightarrow \text{Sets}$ be the functor that maps a scheme S to the set of isomorphism classes of families of n -pointed curves over S , and takes a map $f : S \rightarrow T$ and maps it to the pullback of families:

$$\begin{array}{ccc} \pi : C \rightarrow T & & \tilde{\pi} : C \times_T S \rightarrow S \\ \sigma_i : T \rightarrow C & \mapsto & \tilde{\sigma}_i : S \rightarrow C \end{array}$$

As discussed in Section 6.2, if $\mathcal{M}_{0,n}$ is represented by a scheme then we call that scheme a fine moduli space for $\mathcal{M}_{0,n}$.

Proposition 3.4 (See [KV07]). *For $n \geq 3$ $\mathcal{M}_{0,n}$ admits a fine moduli space, denoted by $M_{0,n}$.*

Example 3.5. For $n = 3$, $\mathcal{M}_{0,3}(\mathbb{C})$ consists of a single 3-pointed curve $\mathbb{P}^1$. Indeed, given any three distinct points in $\mathbb{P}^1$ we can find an automorphism of $\mathbb{P}^1$ that maps those three points to any other three distinct points. Our fine moduli space $M_{0,3}$ consists of a single point.

Example 3.6. We discuss $M_{0,4}$. Given 3 points in $\mathbb{P}^1$ we can find an automorphism that maps those three points to $(0, 1, \infty)$. Thus, given a 4-pointed rational curve $(p_1, p_2, p_3, p_4; \mathbb{P}^1)$ we can apply an automorphism $\lambda : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ and get the 4-pointed curve $(1, 0, \infty, \lambda(p_4); \mathbb{P}^1)$. The number $\lambda(p_4)$ is called the cross-ratio. Explicitly, the cross-ratio given by

$$\lambda(p_4) = \frac{(p_3 - p_1)(p_4 - p_2)}{(p_3 - p_2)(p_4 - p_1)}.$$

Hence to specify a 4-pointed rational curve up to isomorphism we just need to give the cross ratio $\lambda(p_4) \in \mathbb{P}^1 - \{0, 1, \infty\}$. Therefore we expect that

$$M_{0,4} \simeq \mathbb{P}^1 - \{0, 1, \infty\};$$

this is in fact the case. More generally, given n ordered points in $\mathbb{P}^1$ we can take the first three to $(0, 1, \infty)$. We get a cross-ratio for each of the remaining $n - 3$ points. Hence, $M_{0,n}$ consists of the data of $n - 3$ distinct cross ratios. In other words, we expect $M_{0,n}$ to be the product of $n - 3$ copies of $M_{0,4}$ minus repeating coordinates; this is indeed the case.

Note that in the example above we can see that $M_{0,n}$ is not proper. Intuitively, we can also see this by noting that a one parameter family of two marked points approaching each other does not have a limit in $M_{0,n}$. As such, as a first attempt to compactify $M_{0,n}$ we may allow for points to collide. However, if we naively add these curves we end up with a non-separated space. We illustrate this heuristically by considering two distinct limits to the same one-parameter family. Consider two families of 4-marked curves given by

$$(0, 1, \infty, t; \mathbb{P}^1) \quad \text{and} \quad (0, t^{-1}, \infty, 1; \mathbb{P}^1).$$

We may scale the points in the second family by t and realize both families have the same cross ratio: t . Thus the two families should be isomorphic. However, their limiting curves are respectively

$$(0, 1, \infty, \infty; \mathbb{P}^1) \quad \text{and} \quad (0, 0, \infty, 1; \mathbb{P}^1),$$

but these two configurations are not projectively equivalent. A solution to this problem proposed by Deligne and Mumford is to consider the notion of a stable curve.¹

¹Alternatively, points can be allowed to collide if we impose additional conditions. An example of this is the Losev-Manin compactification of $M_{0,n}$ [LM00]

Definition 3.7. A *stable n -pointed curve* C is a projective, connected, reduced, genus 0 curve with at worst nodal singularities, with n distinct non-singular marked points, such that for every irreducible component $E \simeq \mathbb{P}^1$ we have that E has at least three *special points*. We call a point a *special point* if it is either a singularity or a marked point.

We reframe our original moduli problem in terms of stable n -pointed curves. Families over S are given by a flat proper morphism $C \rightarrow S$, with n disjoint sections $S \rightarrow C$ that lands on the smooth locus of C , such that every geometric fiber is a stable curve. We define the functor $\overline{\mathcal{M}}_{0,n}$ by mapping S to isomorphism classes of families of stable n -pointed curves and taking morphism $S \rightarrow T$ to the pullback maps defined in Definition 3.3.

Theorem 3.8. $\overline{\mathcal{M}}_{0,n}$ admits a projective fine moduli space. We denote the fine moduli space as $\overline{M}_{0,n}$. Moreover, it contains $M_{0,n}$ as an open subscheme.

$\overline{M}_{0,n}$ is known as the Deligne-Mumford compactification of $M_{0,n}$. This compactification resolves our earlier problem of two marked points approaching each other. The idea is that in the limit as two marked points approach each other, a new irreducible component, containing the two approaching points, is created. To illustrate this we revisit our previous example.

Example 3.9. Recall our two families

$$(0, 1, \infty, t; \mathbb{P}^1) \quad \text{and} \quad (0, t^{-1}, \infty, 1; \mathbb{P}^1).$$

See Figure 1 for a diagram of their corresponding limits.

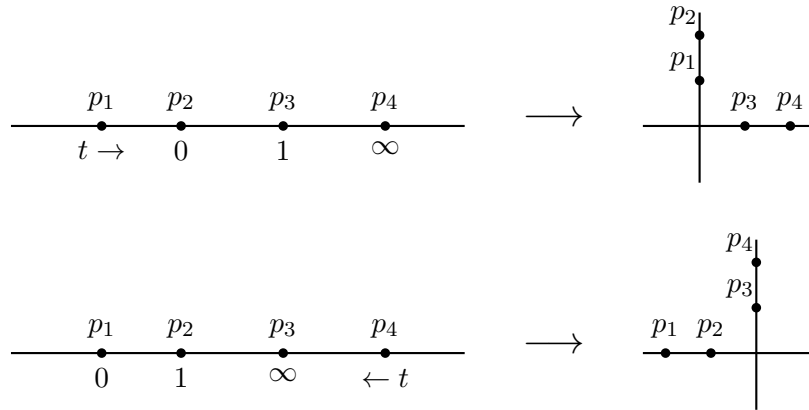


Figure 1: Diagrams depicting the stable nodal curve limits in $\overline{\mathcal{M}}_{0,4}$ of the families $(0, 1, \infty, t; \mathbb{P}^1)$ and $(0, t^{-1}, \infty, 1; \mathbb{P}^1)$. Unlike our previous example, these two limits are now projectively equivalent.

Remark 3.10. As we discussed in Appendix A, in order to establish that $M_{0,n}$ and $\overline{M}_{0,n}$ are fine moduli spaces we must construct their respective universal families, $U_{0,n}$ and $\overline{U}_{0,n}$. While the universal families play an important role in the construction of $\overline{M}_{0,n}$, we will omit a discussion on universal families since they do not play a central role in our discussion of the moduli of maps.

In the following section we will generalize the idea of the moduli of curves by considering the moduli space of maps and its compactification by Kontsevich.

3.2 Moduli of Maps

We move from considering the moduli of n -pointed curves onto studying the moduli of maps from an n -pointed curve to a target space X . To guarantee the existence of the moduli spaces discussed in this section we assume X is projective.

Definition 3.11. Let S be a scheme. A family of n -pointed maps over S consists of

- A flat projective map $\pi : C \rightarrow S$.
- n sections $\sigma_i : S \rightarrow C$, such that each geometric fiber $C_{\overline{s}}$ is an n -marked curve.
- A map $\mu : C \rightarrow X$.

Definition 3.12. Two families $X \leftarrow C \rightarrow S$, $X \leftarrow C' \rightarrow S$ are isomorphic if there exists an isomorphism $C \simeq C'$ compatible with all the data. That is to say, the following diagrams commute:

$$\begin{array}{ccc} C & \xrightarrow{\cong} & C' \\ \swarrow \sigma_i & & \nwarrow \sigma'_i \\ & S & \end{array} \qquad \begin{array}{ccc} C & \xrightarrow{\cong} & C' \\ \searrow & & \swarrow \\ & X & \end{array} .$$

We now have the main ingredients to define a moduli functor. We add an additional parameter to our moduli functor, the degree of the curve.²

Definition 3.13. Let C be curve, and $\beta \in A_1(X)$. We say $\mu : C \rightarrow X$ represents the class β if $\mu_*[C] = \beta$. We say a family $C \rightarrow S$ represents β if over each geometric fiber the map $C_{\overline{s}} \rightarrow X$ represents β .

Definition 3.14. Let $\mathcal{M}_{0,n}(X, \beta) : \text{Sch}^{op} \rightarrow \text{Sets}$ be the functor that takes a scheme S to the set of isomorphism classes of families of n -pointed maps over

²See Example 3.16 for why we call it degree

S such that over each geometric fiber the map $C_{\bar{s}} \rightarrow X$ represents the class β . On morphisms, the functor $\mathcal{M}_{0,n}(X, \beta) : \text{Sch}^{op} \rightarrow \text{Set}$ takes a morphism $f : S \rightarrow T$ and maps it to the pullback of families:

$$\begin{array}{ccc} \pi : C \rightarrow T & & \tilde{\pi} : C \times_T S \rightarrow S \\ \sigma_i : T \rightarrow C & \mapsto & \tilde{\sigma}_i : S \rightarrow C \\ \mu : C \rightarrow X & & \tilde{\mu} : C \times_T S \rightarrow X \end{array} \quad .$$

Proposition 3.15. $\mathcal{M}_{0,n}(X, \beta)$ admits a fine moduli space for $n \geq 3$.

Proof. We defer a sketch of the proof to Remark 3.24. □

We can provide an explicit description of this space for $X = \mathbb{P}^r$.

Example 3.16. Let $X = \mathbb{P}^r$. The chow group $A_1(\mathbb{P}^r)$ is generated by the class of the line, so every $\beta \in A_1(\mathbb{P}^r)$ can be written as $\beta = d[L]$. We call d the degree of the map. A degree d map $\mathbb{P}^1 \rightarrow \mathbb{P}^r$ is given in coordinates by

$$[x : y] \mapsto [p_0(x, y) : \dots : p_r(x, y)].$$

Where the $p_i(x, y)$ are a collection of $r + 1$ degree d homogeneous polynomials that do not simultaneously vanish. Therefore, choice of these polynomials, up to a constant factor, determines a map $\mathbb{P}^1 \rightarrow \mathbb{P}^r$. This choice determines a Zariski open set of $U \subset \mathbb{P}((\bigoplus_{i=0}^r \text{Sym}^d(\mathbb{C}^2)))$. Once we have specified the map, we are left with the choice of placing n marks on the $\mathbb{P}^1$ source. The choices for placing n marks on $\mathbb{P}^1$ are parametrized by $M_{0,n}$. A good guess for our fine moduli space would then be $M_{0,n}(X, d) \simeq M_{0,n} \times U$; this turns out to be correct. See [KV07] for a proof and a description of the universal family.

Kontsevich generalized the Deligne-Mumford compactification of $M_{0,n}$ to n -pointed maps by introducing the notion of a stable maps.

Definition 3.17. Let S be a scheme. A family of n -pointed stable maps over S consists of

- A map $\mu : C \rightarrow X$.
- A flat projective map $\pi : C \rightarrow S$.
- n sections $\sigma_i : S \rightarrow C$, landing in the smooth locus of C .

The map $\pi : C \rightarrow S$ should satisfy that over each geometric fiber, $C_{\bar{s}}$ is projective, connected, reduced, genus 0 curve with at most nodal singularities.

Moreover, over each irreducible component $E \simeq \mathbb{P}^1 \subset C_{\bar{s}}$, if E is mapped to a point under restriction $\pi|_E: E \rightarrow X$, then E must have at least three *special points*.

Remark 3.18. We framed the moduli of curves and maps as a functors $\text{Sch}^{op} \rightarrow \text{Sets}$. In what follows we must replace the category Sets with the category of groupoids Gpd . For $n \geq 3$, a stable n -pointed map $\mathbb{P}^1 \rightarrow X$ has possibly non-trivial, but finitely many, automorphisms. In fact, the stability condition is equivalent to demanding families have finitely many automorphisms [FP97]. By switching from Sets to Gpd we keep track of the data of automorphisms of stable maps. Indeed, we could have framed the moduli functor in this manner from the start. In the case of curves, stable curves, and maps, the corresponding automorphism groups were trivial. There was no information lost by using the composition $\text{Sch}^{op} \rightarrow \text{Gpd} \rightarrow \text{Sets}$; the functor $\text{Gpd} \rightarrow \text{Sets}$ maps a groupoid to the set of its connected components. The “category” Gpd is actually a 2-category; therefore our moduli functor should be replaced by a *pseudofunctor* between 2-categories. We omit the technicalities of defining pseudofunctors and simply define $\overline{\mathcal{M}}_{0,n}(X, \beta)$ on objects. For a discussion of 2-functors see [nLa26].

Definition 3.19. Let $\overline{\mathcal{M}}_{0,n}(X, \beta) : \text{Sch}^{op} \rightarrow \text{Gpd}$ map a scheme S to the groupoid with families of n -pointed stable maps as its objects and isomorphisms of families as morphisms.

Unlike our previous examples, this moduli functor will not admit a fine moduli space. Heuristically we should not expect a fine moduli space due to the presence of non-trivial automorphisms of families. In Example 3.20 we illustrate a non-trivial automorphism of a stable curve with target $\mathbb{P}^1$.

Example 3.20. Let C be the reducible nodal curve obtained by gluing two copies of $\mathbb{P}^1$ along 0. Denote the irreducible components of C by C_1 and C_2 . We turn C into a 3-pointed curve by choosing two points, $p_1, p_2 \in C_1$, and the point $p_3 = \infty \in C_2$. Consider the stable map $f : C \rightarrow \mathbb{P}^1$ that collapses C_1 onto 0 and maps C_2 via the double cover $f([x : y]) = [x^2 : y^2]$ (See Figure 2).

The map $f : C \rightarrow \mathbb{P}^1$ admits an automorphism $\phi : C \rightarrow C$, defined as the identity on C_1 and by $[x : y] \mapsto [-x : y]$ on C_2 . Because this automorphism of C fixes the marked points p_1, p_2 , and $\infty \in C_2$, it is a valid automorphism of the 3-pointed stable map $f : C \rightarrow \mathbb{P}^1$.

Despite $\overline{\mathcal{M}}_{0,n}(X, \beta)$ not admitting a fine moduli space, it does admit a coarse moduli space when X is a projective variety.

Theorem 3.21 (Theorem 2 [FP97]). *Let X be a projective variety and $n \geq 3$*

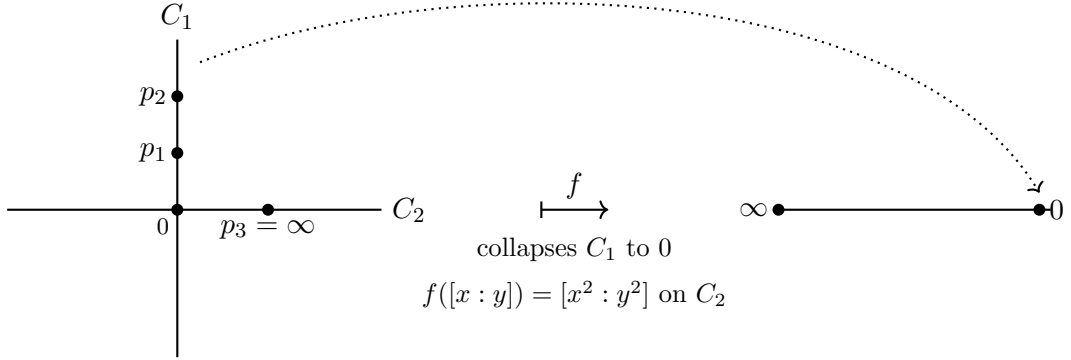


Figure 2: Illustration of 3-pointed stable map $f : C \rightarrow \mathbb{P}^1$.

1. $\overline{\mathcal{M}}_{0,n}(X, \beta)$ admits a projective coarse moduli space, denoted by $\overline{M}_{0,n}(X, \beta)$.
2. Moreover, when $X = G/P$ is a homogeneous variety, $\overline{M}_{0,n}(X, \beta)$ is a normal projective variety of pure dimension

$$\dim(X) + \int_{\beta} c_1(T_X) + n - 3.^3$$

The term $c_1(T_X)$ stands for the first Chern class of the tangent bundle of X . Topologically, Chern classes of a vector bundle E on X are classes in cohomology $c_i(E) \in A^i(X)$. These come from pulling back the special Schubert classes in the cohomology of the infinite Grassmannian $\text{Gr}(k, \infty)$ along the classifying map of the bundle. For a discussion of Chern classes in intersection theory see [Ful84], and for a topological perspective see [Zak26]. The first chern class of the universal quotient bundle Q over the Grassmannian can be described explicitly as the pullback of hyperplane class $\xi \in A^1(\mathbb{P}^r)$ along the Plücker embedding. We use $c_1(Q)$ in Example 3.22 to compute the integral in Theorem 3.21.

Example 3.22. Choose the generator $[L] \in A_1(X)$, such that the pushforward of $[L]$ under the Plücker embedding $\iota : \text{Gr}(k, n) \rightarrow \mathbb{P}^r$ gives us the line class of $\mathbb{P}^r$ $\iota_*[L] \in A_1(\mathbb{P}^r)$. Every $\beta \in A_1(X)$ can then be written as $\beta = d[L]$; we call d the degree of the map. As an application of part two of the theorem above, let us compute the dimension of $\overline{M}_{0,3}(X, d)$. The theorem says

$$(4) \quad \dim(\overline{M}_{0,3}(\text{Gr}(k, n), d)) = \int_{\beta} c_1(T_X) + \dim \text{Gr}(k, n).$$

³This is the “expected dimension”. In more general settings, $\dim(X)$ does not necessarily match the expected dimension.

The dimension of the Grassmannian is known to be $k(n-k)$. Consider the isomorphism $T \operatorname{Gr}(k, n) \simeq S^\vee \otimes Q$, where S is the tautological bundle and Q its quotient bundle (Chapter 2, Exercise 15, [KV07]). By [Stacks, Tag 0FA6]

$$(5) \quad c_1(TX) = (n-k) c_1(S^\vee) + k c_1(Q).$$

We can relate the tautological bundle to the quotient bundle via the following short exact sequence:

$$0 \longrightarrow S \longrightarrow \mathcal{O}_{\operatorname{Gr}(k, n)}^{\oplus n} \longrightarrow Q \longrightarrow 0.$$

It follows that $c_1(\mathcal{O}_{\operatorname{Gr}(k, n)}^{\oplus n}) = c_1(S) + c_1(Q)$ ([Har77] Appendix A, C3). The Chern classes of the trivial bundle vanish so

$$c_1(Q) = -c_1(S) = c_1(S^\vee).$$

From (5) it follows that $c_1(TX) = n c_1(Q)$. The fact that $c_1(Q)$ is the pullback of the hyperplane class $\xi \in A^1(\mathbb{P}^r)$, together with the fact that $[L]$ pushes forward to the line class, allows us to push everything to the geometry of $\mathbb{P}^r$ using the projection formula (17) discussed in Appendix B.

$$\int_{\beta} c_1(Q) = \beta \cap c_1(Q) = d[L] \cap \iota^* \xi = d(\iota_*[L] \cap \xi) = d.$$

Plugging our results back in (4), we conclude

$$\dim(\overline{M}_{0,3}(\operatorname{Gr}(k, n))) = nd + k(n-k).$$

Example 3.23. Let $\beta = 0$. Then by the stability condition the source of every stable map $\mu : C \rightarrow X$ must be a stable curve. Therefore, the data in a degree 0 stable map is a point in X and a stable curve. Indeed we have that

$$\overline{M}_{0,n}(X, 0) \simeq \overline{M}_{0,n} \times X.$$

In the next section we will restrict ourselves to the case where $X = G/P$ is a homogeneous space. We define Gromov-Witten invariants using the coarse moduli space $\overline{M}_{0,n}(X, \beta)$. A key ingredient are the evaluation maps $\rho_i : \overline{M}_{0,n}(X, \beta) \rightarrow X$. These are given by taking a stable map $\mu : C \rightarrow X$ and mapping it to the image under μ of the i -th marked point of C . The evaluations maps are well-defined since we require an isomorphism of families to commute with the sections; thus the images of marked points are preserved under isomorphism of

families.

Remark 3.24. The proof uses basic concepts in the theory of algebraic stacks that are beyond the scope of this thesis. We will sketch out the main ideas. Let $\overline{\mathcal{M}}$ denote $\overline{\mathcal{M}}_{0,n}(X, \beta)$ as an algebraic stack. The moduli space of stable maps from smooth source curves $\mathcal{M}_{0,n}(X, \beta)$ is an open sub-stack of $\overline{\mathcal{M}}$ (See [Stacks, Tag 0DZY]). The morphism $\overline{\mathcal{M}} \rightarrow \overline{\mathcal{M}}_{0,n}(X, \beta)$ is an isomorphism on this open substack; the isomorphism follows from the fact that stable maps with smooth source curves have no non-trivial automorphisms for $n \geq 3$. We can then check that we get a universal family for $\mathcal{M}_{0,n}(X, \beta)$ through the universal family (as a moduli stack) of $\overline{\mathcal{M}}$.

4 Quantum Cohomology

4.1 Gromov-Witten Invariants

Gromov-Witten invariants intuitively count the number of curves of a fixed degree passing through a collection of subvarieties of X . In some cases, this intuition is precise, and we say the corresponding Gromov-Witten invariant is enumerative, as in Lemma 4.2. In this section, we define Gromov-Witten invariants, prove some important properties about them, and turn our discussion to quantum cohomology. Throughout this section, X will be a homogeneous space, so that the results of Theorem 3.21 apply.

Definition 4.1. Let $\gamma_1, \dots, \gamma_n \in A^*(X)$, and $\beta \in A_1(X)$. We define the Gromov-Witten invariant $\langle \gamma_1, \dots, \gamma_n \rangle_\beta$:

$$\langle \gamma_1, \dots, \gamma_n \rangle_\beta = \int_{\overline{\mathcal{M}}_{0,n}(X, \beta)} \rho_1^*(\gamma_1) \cup \dots \cup \rho_n^*(\gamma_n).$$

Let us unpack this definition. In genus 0, the evaluation maps ρ_i , discussed at the end of Section 3, are flat. Therefore, if the γ_i are the Poincare duals of subvarieties $[\Gamma_i] \in A_*(X)$, we may compute the pullback simply as a flat pullback, as discussed in Appendix B. Using the flat pullback and the duality between the intersection product and the cup product, we can rewrite the integral as

$$(6) \quad \int_{\overline{\mathcal{M}}_{0,n}(X, \beta)} \rho_1^*(\gamma_1) \cdots \rho_n^*(\gamma_n) = \int_{\overline{\mathcal{M}}_{0,n}(X, \beta)} [\rho_1^{-1}(\Gamma_1)] \cdot [\rho_2^{-1}(\Gamma_2)] \cdots [\rho_n^{-1}(\Gamma_n)].$$

Theorem 3.21 tells us $\overline{\mathcal{M}}_{0,n}(X, \beta)$ is a normal projective variety; for the sake of illustration let us assume that it is in fact smooth so that we can apply Lemma

7.4.

We use Lemma 7.4 to translate $\rho^{-1}(\Gamma_1), \dots, \rho^{-1}(\Gamma_n)$ to $\rho^{-1}(\Gamma_1)', \dots, \rho^{-1}(\Gamma_n)'$ so that these preimages intersect transversely. Substituting back into (6) we get

$$\langle \gamma_1, \dots, \gamma_n \rangle_\beta = \int_{\overline{M}_{0,n}(X, \beta)} [\rho_1^{-1}(\Gamma_1)' \cap \rho_2^{-1}(\Gamma_2)' \cap \dots \cap \rho_n^{-1}(\Gamma_n)'] .$$

The term inside the integral represents the intersection of n subvarieties in $\overline{M}_{0,n}(X, \beta)$. The integral above is only non-zero when the intersection $\rho_1^{-1}(\Gamma_1)' \cap \rho_2^{-1}(\Gamma_2)' \cap \dots \cap \rho_n^{-1}(\Gamma_n)'$ is a finite collection of points. The points correspond to n -pointed stable maps whose i -th marked point lies in the subvariety Γ_i . The integral then computes the number of such n -pointed stable maps. Ideally, this count would be supported in $M_{0,n}(X, \beta)$, so that the integral counts stable maps with smooth source. While we assumed $\overline{M}_{0,n}(X, \beta)$ was smooth, the assumption is not needed to establish the integral as a count of stable maps intersecting subvarieties at the marked points; this is the content of Lemma 14 in [FP97].

Lemma 4.2 (Lemma 14, [FP97]). *Let $\Gamma_1, \dots, \Gamma_n$ be pure dimensional subvarieties of $X = G/P$ such that*

$$\sum_{i=1}^n \text{codim}(\Gamma_i) = \dim(\overline{M}_{0,n}(X, \beta)) .$$

Let $g_1, \dots, g_n \in G$ be general elements. Then, the scheme theoretic intersection

$$\rho_1^{-1}(g_1\Gamma_1) \cap \dots \cap \rho_n^{-1}(g_n\Gamma_n)$$

is a finite number of reduced points supported in $M_{0,n}(X, \beta)$ and

$$\langle \gamma_1, \dots, \gamma_n \rangle_\beta = |\rho_1^{-1}(g_1\Gamma_1) \cap \dots \cap \rho_n^{-1}(g_n\Gamma_n)| .$$

Remark 4.3. Gromov-Witten invariants may be defined for arbitrary genus g in a similar way as Definition 4.1. However, in general we would no longer integrate over the fundamental class of coarse moduli space $[\overline{M}_{0,n}(X, \beta)]$. We instead have to define a “virtual fundamental class” $[\overline{M}_{0,n}(X, \beta)]^{\text{vir}}$ whose dimension agrees with the expected dimension found in Theorem 3.21. For more on the virtual fundamental class see [BF97].

4.2 Two Properties of Gromov-Witten invariants

We prove two important properties of Gromov-Witten invariants, which will be helpful when we study quantum cohomology. The first property says that degree zero Gromov-Witten invariants compute triple intersections.

Lemma 4.4. *Let $\beta = 0$, and $n \geq 3$. Take $\gamma_1, \dots, \gamma_n \in A^*(X)$, then*

$$\langle \gamma_1, \dots, \gamma_n \rangle_0 = \begin{cases} 0 & \text{for } n > 3 \\ \int_X \gamma_1 \cup \gamma_2 \cup \gamma_3 & \text{for } n = 3 \end{cases}.$$

Proof. In Example 3.23 we saw that $\overline{M}_{0,n}(X, 0) \simeq \overline{M}_{0,n} \times X$. Hence

$$\langle \gamma_1, \dots, \gamma_n \rangle_0 = \int_{\overline{M}_{0,n} \times X} \rho_1^*(\gamma_1) \cup \dots \cup \rho_n^*(\gamma_n).$$

Because the stable maps are constant, the evaluation maps all coincide with the projection map $\pi : \overline{M}_{0,n} \times X \rightarrow X$. Consequently, we can collect the classes γ_i into a single term,

$$\int_{\overline{M}_{0,n} \times X} \rho_1^*(\gamma_1) \cup \dots \cup \rho_n^*(\gamma_n) = \int_{\overline{M}_{0,n} \times X} \pi^*(\gamma_1 \cup \dots \cup \gamma_n).$$

By the projection formula (17) discussed in Appendix B, it follows that

$$\langle \gamma_1, \dots, \gamma_n \rangle_0 = \int_{\pi_*[\overline{M}_{0,n} \times X]} \gamma_1 \cup \dots \cup \gamma_n.$$

The class $\pi_*[\overline{M}_{0,n} \times X]$ is only non-zero when it matches the dimension of X , as per (16) in Appendix B. The dimensions only match when $n = 3$, since $\overline{M}_{0,3} = \{*\}$ as seen in Example 3.5. Lastly, we note that $\pi_*[\overline{M}_{0,3} \times X] = [X]$. We conclude our desired result

$$\langle \gamma_1, \dots, \gamma_n \rangle_0 = \begin{cases} 0 & \text{for } n > 3 \\ \int_X \gamma_1 \cup \gamma_2 \cup \gamma_3 & \text{for } n = 3 \end{cases}.$$

If $\gamma_1, \dots, \gamma_3$ are the Poincare duals of $\Gamma_1, \Gamma_2, \Gamma_3$, and the intersection $\Gamma_1 \cap \Gamma_2 \cap \Gamma_3$ is finite, then the integral computes the number of points in the intersection $\Gamma_1 \cap \Gamma_2 \cap \Gamma_3$. $\square$

The second property is the divisor equation, which provides a simple formula for computing Gromov-Witten invariants involving divisor classes.

Lemma 4.5. *Let $\gamma_1 \in A^1(X)$, and $\beta \neq 0$, then*

$$\langle \gamma_1, \dots, \gamma_n \rangle_\beta = \left(\int_\beta \gamma_1 \right) \langle \gamma_2, \dots, \gamma_n \rangle_\beta .$$

Proof. Let $\mu : C \rightarrow X$ be a stable map with $\mu_*[C] = \beta$. If $\gamma_1 \in A^1(X)$ is the class of a divisor Γ_1 , then the intersection number $\int_\beta \gamma_1$ is exactly the number of points on C that map into Γ_1 . Lemma 4.2 gives an enumerative interpretation for Gromov-Witten invariants involving the divisor class γ_1 . For any given $(n-1)$ -pointed map $C \rightarrow X$ sending $p_2, \dots, p_n$ to $\Gamma_2, \dots, \Gamma_n$, we can get exactly $\int_\beta \gamma_1$ distinct n -pointed maps, each corresponding to a valid choice for the new marked point $p_1 \in C$. □

4.3 Quantum Cohomology

The quantum cohomology ring is defined using the genus 0 Gromov-Witten invariants. Let $\gamma_0 = 1 \in A^0(X)$, and $\gamma_1, \dots, \gamma_p$ be a $\mathbb{Z}$ basis for $A^1(X)$, and $\gamma_{p+1}, \dots, \gamma_m$ be a basis for the rest of $A^*(X)$. Let g_{ij} denote the intersection pairing

$$g_{ij} = \int_X \gamma_i \cup \gamma_j .$$

We let g^{ij} be the entries of the matrix inverse of the $m \times m$ matrix g_{ij} . There is a relation between the usual product on $A^*(X)$ and the degree 0 Gromov-Witten invariants given by

$$(7) \quad \gamma_i \cup \gamma_j = \sum_{e,f=0}^m \langle \gamma_i, \gamma_j, \gamma_e \rangle_0 g^{ef} \gamma_f .$$

The expression above is a consequence of applying Lemma 4.4 to (20) discussed in Appendix B. The idea of quantum cohomology is to take (7) and allow for contributions to the product coming from non-zero β . Our goal is to write the quantum product in the form

$$(8) \quad \gamma_i \star \gamma_j = \sum_{e,f=0}^m \Phi_{ije} g^{ef} \gamma_f .$$

The coefficient Φ_{ije} should contain the data of all possible β . The key idea coming from physics is to encode Φ_{ije} through the partial derivatives of a

“potential function” which we call Φ . Quantum cohomology is defined using the following potential function

$$\Phi(y_0, \dots, y_m) = \sum_{n_0 + \dots + n_m \geq 3} \sum_{\beta} \langle \gamma_0^{n_0}, \dots, \gamma_m^{n_m} \rangle_{\beta} \frac{y_0^{n_0}}{n_0!} \dots \frac{y_m^{n_m}}{n_m!}.$$

Here by γ^n we mean that γ is included n times. Define Φ_{ijk} to be the partial derivatives

$$\Phi_{ijk} = \frac{\partial \Phi}{\partial y_i \partial y_j \partial y_k}.$$

Let $\gamma = \sum y_i \gamma_i$. After computing the triple derivatives we arrive at

$$\Phi_{ijk} = \sum_{n \geq 0} \sum_{\beta} \frac{1}{n!} \langle \gamma^n, \gamma_i, \gamma_j, \gamma_k \rangle_{\beta}.$$

Let $\mathbb{Q}[[y]] := \mathbb{Q}[[y_0, \dots, y_m]]$. We define the quantum product on the $\mathbb{Q}[[y]]$ -module $A^*(X) \otimes \mathbb{Q}[[y]]$, by (8) on basis elements $\gamma_i, \gamma_j \in A^*(X)$ and extending the product to the rest of $A^*(X) \otimes \mathbb{Q}[[y]]$ by extending $\mathbb{Q}[[y]]$ linearly. It is an important theorem that the product (8) gives $A^*(X) \otimes \mathbb{Q}[[y]]$ the structure of a *commutative, associative*, $\mathbb{Q}[[y]]$ -algebra with unit γ_0 . We can easily see that commutativity follows from the commutativity of the cup product. For associativity, and to show γ_0 is a unit, we refer to [FP97] for a complete proof.

4.4 Small Quantum Cohomology

In practice, the quantum cohomology ring is very hard to compute. By restricting to a simpler version of quantum cohomology, called small quantum cohomology, we get a more computable ring. The small quantum cohomology ring provides a deformation of $A^*(X)$ that depends solely on 3-point Gromov-Witten invariants. We will start by defining the product, and then we will show it matches with our original definition of quantum cohomology when restricted to divisor classes.

Let $q_1, \dots, q_p$ denote formal parameters, each corresponding to a divisor class $\{\gamma_i\}_{i=1}^p$ in the fixed basis for $A^1(X)$. Let $\mathbb{Z}[q] = \mathbb{Z}[q_1, \dots, q_p]$. We define the small quantum cohomology ring as a $\mathbb{Z}[q]$ -algebra with underlying module structure $A^*(X) \otimes \mathbb{Z}[q]$, and product given by

$$(9) \quad \gamma_i \star \gamma_j = \gamma_i \cdot \gamma_j + \sum_{e,f=0}^m \left(\sum_{\beta \neq 0} \langle \gamma_i, \gamma_j, \gamma_e \rangle_{\beta} q_1^{\int_{\beta} \gamma_1} \dots q_p^{\int_{\beta} \gamma_p} \right) g^{ef} \gamma_f.$$

From the presentation above it is clear that when we set the deformation parameters q_i to zero we recover the classical product on A^*X .

Let us see how to recover the small quantum cohomology from our original definition of quantum cohomology. Recall the expression for Φ_{ijk} :

$$\Phi_{ijk} = \sum_{n \geq 0} \sum_{\beta} \frac{1}{n!} \langle \gamma^n, \gamma_i, \gamma_j, \gamma_k \rangle_{\beta} .$$

The important transition from ordinary quantum cohomology to small quantum cohomology is that the formal parameters now only come from divisor classes. That is, $\gamma = \sum_{i=1}^p y_i \gamma_i$, where $\gamma_i, \dots, \gamma_p$ are our divisor classes. We can split Φ_{ijk} into two parts as

$$\Phi_{ijk} = \sum_{n \geq 0} \frac{1}{n!} \langle \gamma^n, \gamma_i, \gamma_j, \gamma_k \rangle_0 + \sum_{n \geq 0} \sum_{\beta \neq 0} \frac{1}{n!} \langle \gamma^n, \gamma_i, \gamma_j, \gamma_k \rangle_{\beta} .$$

Applying Lemma 4.4 to the first expression, we find

$$\sum_{n \geq 0} \frac{1}{n!} \langle \gamma^n, \gamma_i, \gamma_j, \gamma_k \rangle_0 = \langle \gamma_i, \gamma_j, \gamma_k \rangle_0 .$$

We will call the $\beta = 0$ part the classical part of the potential, and denote it as Φ_{class} . If we substitute Φ_{class} into (8), then by the classical product formula (20) we found in Appendix B, we can write

$$\gamma_i \star \gamma_j = \gamma_i \cup \gamma_j + \sum_{e,f=0}^m \left(\sum_{n \geq 0} \frac{1}{n!} \sum_{\beta \neq 0} \langle \gamma^n, \gamma_i, \gamma_j, \gamma_e \rangle_{\beta} \right) g^{ef} \gamma_f .$$

Since γ is now a sum of divisor classes, for $\beta \neq 0$ we can use the divisor equation Lemma 4.5 to pull out each γ . We are left with

$$\begin{aligned} \sum_{n \geq 0} \frac{1}{n!} \sum_{\beta \neq 0} \langle \gamma^n, \gamma_i, \gamma_j, \gamma_k \rangle_{\beta} &= \sum_{n \geq 0} \frac{1}{n!} \left(\sum_{l=1}^n y_l \int_{\beta} \gamma_l \right)^n \langle \gamma_i, \gamma_j, \gamma_k \rangle_{\beta} \\ &= e^{y_1 \int_{\beta} \gamma_1} \dots e^{y_p \int_{\beta} \gamma_p} \langle \gamma_i, \gamma_j, \gamma_k \rangle_{\beta} . \end{aligned}$$

If we set $q_i = e^{y_i}$, then we can rewrite the quantum product as

$$\gamma_i \star \gamma_j = \gamma_i \cdot \gamma_j + \sum_{e,f=1}^m \left(\sum_{\beta \neq 0} \langle \gamma_i, \gamma_j, \gamma_e \rangle_{\beta} q_1^{\int_{\beta} \gamma_1} \dots q_p^{\int_{\beta} \gamma_p} \right) g^{ef} \gamma_f .$$

In the following sections, when we say “quantum cohomology” we will be

referring exclusively to the small quantum cohomology, and thus will be denoted as $QH^*(X)$. A great deal of work has been done in computing the quantum cohomology for flag varieties, starting with the work of Bertram in the case of the Grassmannian [Ber97]. This was later expanded to the case of full flag varieties by the work of [FGP97], [Cio99b], and [Kim96]. The case of partial flags was subsequently worked out by [Cio99a] as an extension of the methods used in [FGP97] and [Cio99b]. In the following section, we will focus on the simplest case, the quantum cohomology of the Grassmannian. We will discuss the quantum analogues of the Pieri and Giambelli formulas found by [Ber97]. Before we proceed, we end this section by computing the small quantum cohomology of $X = \mathbb{P}^r$.

Example 4.6. $QH^*(\mathbb{P}^r) \simeq \mathbb{Z}[h, q]/(h^{r+1} - q)$, where h is the class of the hyperplane.

Proof. Let $h \in A^1(\mathbb{P}^r)$ denote the hyperplane class. It is well-known that h generates $A^1(\mathbb{P}^r)$ as a $\mathbb{Z}$ -module. Therefore any $\beta \in A^1(\mathbb{P}^r)$ can be written as $\beta = dh$. If β is to represent the class of a curve then we must have $d \geq 0$. Therefore, instead of writing β we will use $d \in \mathbb{Z}_{\geq 0}$ where $\beta = dh$. Furthermore, the class h generates $A^*(\mathbb{P}^r)$ as a ring. In fact, we have the presentation,

$$A^*(\mathbb{P}^r) \simeq \mathbb{Z}[h]/(h^{r+1}) \quad .$$

Our task is to compute the quantum product $h^i \star h^j$ for $i, j \leq r$. The intersection pairing g^{ef} is only non-zero if the sum of the codimensions $e + f = r$. It follows that we can write the quantum product $h^i \star h^j$ as

$$h^i \star h^j = h^{i+j} + \sum_{d=1} \langle h^i, h^j, h^e \rangle_d q^d h^{r-e} \quad .$$

Recall that $\langle h^i, h^j, h^e \rangle_d \neq 0$ only if $i + j + e = \dim(M_{0,3}(\mathbb{P}^r, d)) = r + (r+1)d$. Because $i, j, e \leq r$, the equality above can only happen at $d = 0, 1$. Therefore we consider two cases:

- Consider the case $i + j \leq r$. Since $e \leq r$ we deduce $i + j + e \leq 2r$. In particular, $i + j + e < r + (r+1)$, so it follows that

$$\langle h^i, h^j, h^e \rangle_1 = 0.$$

In this case we conclude $h^i \star h^j = h^{i+j}$.

- Now consider the case $r + 1 \leq i + j \leq 2r$. Then $i + j + e > r$ so

$$\langle h^i, h^j, h^e \rangle_0 = 0.$$

We turn to consider the degree one term $\langle h^i, h^j, h^e \rangle_1$. This term is only non-zero when $i + j + e = 2r + 1$, therefore we set $e = 2r + 1 - (i + j)$. Since $\mathbb{P}^r$ is a homogeneous variety then by Lemma 4.2 we can count $\langle h^i, h^j, h^e \rangle_1$ by looking at stable maps with source $\mathbb{P}^1$. A degree one map $T : \mathbb{P}^1 \rightarrow \mathbb{P}^r$ corresponds to a map

$$T : [x : y] \mapsto [T_0(x, y) : T_1(x, y) : T_2(x, y) : \cdots : T_r(x, y)],$$

where each T_i is a linear form. The map T is induced from a linear map $\tilde{T} : \mathbb{C}^2 \rightarrow \mathbb{C}^{r+1}$. At the level of vector spaces, demanding that the marked points $p_1, p_2, p_3 \in \mathbb{C}^2$ land in the subvarieties corresponding to h^i, h^j, h^e under $T : \mathbb{P}^1 \rightarrow \mathbb{P}^r$, is equivalent to demanding that linearly independent vectors $v_{p_1}, v_{p_2}, v_{p_3} \in \mathbb{C}^2$ land in the subspaces W_i, W_j, W_e of codimensions i, j, e respectively. The subspaces W_i, W_j, W_e correspond to the hyperplane classes h^i, h^j, h^e . The matrix $\tilde{T}$ has $2r + 2$ coefficients. Our goal will be to show that after we impose the constraints of the map $\tilde{T}$, we will be left with only one free coefficient. Each subspace W_i can be defined by the intersection of i hyperplanes in $\mathbb{C}^{r+1}$. Thus requiring $\tilde{T}(v_{p_1}) \in W_i$ imposes i linear constraints on the coefficients of $\tilde{T}$. Assuming these constraints are independent, i.e the subspaces W_i, W_j, W_e are generic, it follows we get $i + j + e = 2r + 1$ linear constraints on the coefficients of $\tilde{T}$. Consequently, $\tilde{T}$ has only one free parameter. Thus $\tilde{T}$ defines a unique map in projective space $T : \mathbb{P}^1 \rightarrow \mathbb{P}^r$. It follows that

$$\langle h^i, h^j, h^e \rangle_1 = 1.$$

In this case we conclude $h^i \star h^j = qh^{i+j-r-1}$.

We conclude that $QH^*(\mathbb{P}^r) \simeq \mathbb{Z}[h, q]/(h^{r+1} - q)$. As a sanity check, see that for $q = 0$ we recover $A^*(\mathbb{P}^r)$. □

5 Quantum Cohomology of the Grassmannian

In this section, we discuss the quantum analogues of the Pieri and Giambelli formulas. These analogues were initially studied by Bertram [Ber97] and later

proved through more elementary means by Buch [Buc03]. The following discussion closely follows Buch's approach. We begin with a brief overview of the classical cohomology of the Grassmannian. From now on let $X = \text{Gr}(k, n)$, the Grassmannian of k -planes in $\mathbb{C}^n$.

5.1 Quantum Cohomology of the Grassmannian

In Section 2 we saw that the cohomology ring of the Grassmannian is generated by the Schubert classes Ω_λ . Hence, we can define the quantum cohomology in the Grassmannian by choosing the Schubert classes as generators of $A^*(X)$. For our degree $\beta \in A_1(X)$, recall from Example 3.22 that β can be uniquely specified by an integer $d \in \mathbb{Z}$, which we call the degree. Thus the coefficients in the quantum product are determined by 3-point Gromov-Witten invariants $\langle \Omega_\lambda, \Omega_\mu, \Omega_\nu \rangle_d$.

These Gromov-Witten invariants count the number of degree d rational curves on $\text{Gr}(k, n)$ that meet the Schubert varieties $\Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet), \Omega_\nu(H_\bullet)$ for general flags $F_\bullet, G_\bullet, H_\bullet$. Further, $A^1(X)$ only has one generator, namely Ω_1 , the partition corresponding to a single box. This tells us that $QH^*(X)$ only has one deformation parameter q . The quantum product is thus given by

$$\Omega_\lambda \star \Omega_\mu = \sum_{\nu, d \geq 0} \langle \Omega_\lambda, \Omega_\mu, \Omega_\nu \rangle_d q^d g^{\gamma^\nu} \Omega_\nu.$$

It is a fact from Schubert calculus that $g^{\gamma^\nu} = 1$ if and only if $\gamma = \nu^\vee$ where $\nu^\vee$ is the complementary partition of ν . We can obtain $\nu^\vee$ by looking at the Young diagram of ν and taking the complement of the diagram in the $k \times (n - k)$ rectangle then rotating that diagram to get a valid partition. As a consequence we can write the quantum product as

$$(10) \quad \Omega_\lambda \star \Omega_\mu = \sum_{\nu, d \geq 0} \langle \Omega_\lambda, \Omega_\mu, \Omega_{\nu^\vee} \rangle_d q^d \Omega_\nu.$$

5.2 Quantum Pieri and Giambelli's Formula

We now present the statements of the Quantum Pieri Formula and the Quantum Giambelli; before providing their proofs in Section 5.4.

Theorem 5.1 (Quantum Pieri Formula, Bertram). *If λ is contained in an $k \times (n - k)$ rectangle and $p \leq n - k$ then*

$$\Omega_p \star \Omega_\lambda = \sum_{\substack{\lambda_i \leq \mu_i \leq \lambda_{i-1} \\ |\mu| = p + |\lambda|}} \Omega_\mu + q \left(\sum_{\substack{0 \leq \lambda_i - 1 \leq \nu_i \leq \lambda_{i-1} - 1 \\ |\nu| = p + |\lambda| - n}} \Omega_\nu \right)$$

Theorem 5.2 (Quantum Giambelli Formula, Bertram). *If the Young diagram of the partition λ is contained in an $k \times (n - k)$ rectangle, then the Schubert class Ω_λ in $QH^*Gr(k, n)$ is given by $\Omega_\lambda = \det(\Omega_{\lambda_i + j - i})_{1 \leq i, j \leq k}$.*

In the Quantum Giambelli formula the determinant is taken using the quantum product. However, in the proof we will see that the quantum products in the determinant are the same as the classical products. Before we move on to a proof, we illustrate the computation of the quantum product through an example.

Example 5.3. Let $n = 6, k = 3$. We will consider $\lambda = (1, 1, 1), \mu = (2, 1, 0)$ as in Example 2.8. Our goal is to compute their quantum product $\Omega_\lambda \star \Omega_\mu$ using the quantum Pieri and Giambelli Formula. First we may use the quantum Giambelli formula on Ω_μ to write it as a product of special Schubert classes.

$$\Omega_{\begin{smallmatrix} \square & \square \end{smallmatrix}} = \det \begin{bmatrix} \Omega_2 & \Omega_3 & \Omega_4 \\ \Omega_0 & \Omega_1 & \Omega_2 \\ \Omega_{-2} & \Omega_{-1} & \Omega_0 \end{bmatrix} = -\Omega_3 + \Omega_2 \star \Omega_1$$

Here we use the fact that $\Omega_0 = 1$, and $\Omega_p = 0$ for $p < 0$ or $p > 3$. By associativity of the quantum product we compute

$$(11) \quad \Omega_\lambda \star \Omega_\mu = -\Omega_\lambda \star \Omega_3 + (\Omega_\lambda \star \Omega_2) \star \Omega_1.$$

Let us compute each term separately, starting with $\Omega_\lambda \star \Omega_3$. The classical term is 0 as computed in Example 2.8. For the q -term we must count the number of ν that can be obtained from the empty partition, $(0, 0, 0) = (1 - 1, 1 - 1, 1 - 1)$, by adding $p - n + |\lambda| = 3 - 3 + 3 = 3$ boxes with no two in the same column. There is only one such partition, namely $\begin{smallmatrix} \square & \square & \square \end{smallmatrix}$. It follows that

$$\Omega_\lambda \star \Omega_3 = \Omega_{\begin{smallmatrix} \square \\ \square \end{smallmatrix}} \star \Omega_{\begin{smallmatrix} \square & \square & \square \end{smallmatrix}} = q\Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}}.$$

Now we turn our attention to $\Omega_\lambda \star \Omega_2$. From Example 2.8, the only partition that contributes to the classical term is $\begin{smallmatrix} \square & \square \end{smallmatrix}$. For the q -term we must count the number of ν that can be obtained from the empty partition, $(0, 0, 0) = (1 - 1, 1 - 1, 1 - 1)$, by adding $p - n + |\lambda| = 2 - 3 + 3 = 2$ boxes with no two in the same column. There is only one such partition, namely $\begin{smallmatrix} \square & \square \end{smallmatrix}$, therefore

$$\Omega_\lambda \star \Omega_2 = \Omega_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} \star \Omega_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} = \Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} + q\Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}}.$$

Now we must compute $\Omega_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} \star \Omega_1$ and $\Omega_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} \star \Omega_1$. By performing the same analysis as before we arrive at the following results:

$$\Omega_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} \star \Omega_1 = \Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} + q\Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} ;$$

$$\Omega_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} \star \Omega_1 = \Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} + q\Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}}.$$

Replacing the expressions we found for $\Omega_\lambda \star \Omega_3$ and $\Omega_\lambda \star \Omega_2 \star \Omega_1$ back in (11) we arrive at

$$\begin{aligned} \Omega_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} \star \Omega_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} &= -q\Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} + \Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} + q(\Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} + q\Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}}), \\ &= \Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}} - q^2\Omega_{\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix}}. \end{aligned}$$

As a sanity check we see that for $q = 0$ we recover the classical answer.

5.3 Preliminaries for Proof

In this section we build up towards proving the quantum Giambelli and quantum Pieri formula as proved by Buch [Buc03]. The proofs in this section rely heavily on Lemma 5.7 which we will state without proof. Let us begin with the following proposition:

Proposition 5.4. $\langle \Omega_\lambda, \Omega_\mu, \Omega_\nu \rangle_d = 0$ for $|\lambda| + |\mu| + |\nu| \neq k(n - k) + dn$.

Proof. Recall $\langle \Omega_\lambda, \Omega_\mu, \Omega_\nu \rangle_d = 0$ if the sum of the co-dimensions of the corresponding varieties is not equal to the dimension of $M_{0,3}(\text{Gr}(k, n), d)$, since the co-dimension of $\Omega_\lambda(F_\bullet)$ is given by $|\lambda|$ this conditions amounts to saying $\langle \Omega_\lambda, \Omega_\mu, \Omega_\nu \rangle_d = 0$ if $|\lambda| + |\mu| + |\nu| \neq \dim(M_{0,3}(\text{Gr}(k, n), d))$. In Example 3.22 we saw $M_{0,3}(\text{Gr}(k, n), d) = k(n - k) + dn$. $\square$

Let us now move onto stating Lemma 5.7.

Definition 5.5. Let C be a rational curve in $\text{Gr}(k, n)$. Define the *span* of C to be the smallest subspace of $\mathbb{C}^n$ containing all the k -dimensional subspaces given by the points of C .

Definition 5.6. For a partition λ and an integer $d > 0$, let $\tilde{\lambda}$ be the partition given by $\tilde{\lambda}_i = \max(\lambda_i - d, 0)$.

Lemma 5.7 (Lemma 2, [Buc03]). *Let $C \subset \text{Gr}(k, n)$ be a rational curve of degree $d \leq n - k$ and let $W \subset \mathbb{C}^n$ be $k + d$ dimensional subspace containing the span of C . If λ is a partition such that $C \cap \Omega_\lambda(F_\bullet) \neq \emptyset$ then W belongs to the Schubert variety $\Omega_{\tilde{\lambda}}(F_\bullet)$ in $\text{Gr}(k + d, n)$.*

The following is an application of Lemma 5.7:

Definition 5.8. Let $\ell(\lambda)$ denote the length of a partition λ , defined as the number of rows in its associated Young diagram.

Proposition 5.9. *Let λ and μ be partitions contained in an $k \times (n - k)$ rectangle such that $\ell(\lambda) + \ell(\mu) \leq k$. Then $\Omega_\lambda \star \Omega_\mu = \Omega_\lambda \cdot \Omega_\mu$.*

Proof. We must show that the contributions to the quantum product of $d > 0$ all vanish. By Proposition 5.4 $\langle \Omega_\lambda, \Omega_\mu, \Omega_\nu \rangle_d \neq 0$ can only happen for ν such that

$$|\lambda| + |\mu| + |\nu| = k(n - k) + nd.$$

Suppose this was the case and $\langle \Omega_\lambda, \Omega_\mu, \Omega_\nu \rangle_d \neq 0$. Then for a choice of general flags $F_\bullet, G_\bullet, H_\bullet$, there is a curve C that intersects $\Omega_\lambda(F_\bullet)$, $\Omega_\mu(G_\bullet)$, and $\Omega_\nu(H_\bullet)$. Consider the case $d \leq n - k$; Lemma 5.7 tells us that for any $k + d$ dimensional subspace W containing the span of C , W is in the intersection

$$\Omega_{\tilde{\lambda}}(F_\bullet) \cap \Omega_{\tilde{\mu}}(G_\bullet) \cap \Omega_{\tilde{\nu}}(H_\bullet) \quad \text{in } \text{Gr}(k + d, n).$$

If this intersection is non-empty then the degree-zero Gromov-Witten invariant of $\text{Gr}(k + d, n)$ corresponding to $\langle \Omega_{\tilde{\lambda}}, \Omega_{\tilde{\mu}}, \Omega_{\tilde{\nu}} \rangle_0$ is non-zero. However, we can argue this invariant is 0 since

$$|\tilde{\lambda}| + |\tilde{\mu}| + |\tilde{\nu}| \geq |\lambda| + |\mu| + |\nu| - 2dk = k(n - k) + d(n - k) - dk > (k + d)(n - k - d).$$

The first inequality comes from considering at most how many boxes can be removed from the Young diagrams of λ, μ, ν . At each row there can be at most d boxes removed. Further, since $\ell(\lambda) + \ell(\mu) \leq k$, we know that λ and μ have at most k rows together. Therefore $|\lambda| + |\mu|$ can decrease at most by dk . Note that ν can be any partition fitting inside a $k \times (n - k)$ rectangle; therefore in the worse case scenario the number of boxes decreases by at most dk . The second inequality follows from $d > 0 \Rightarrow d + k > k$; the steps are as follows:

$$k(n - k) + d(n - k) - dk > (k + d)(n - k) - d(k + d) = (k + d)(n - k - d).$$

The intersection is empty by Proposition 5.4. In particular, assuming the intersection is non-empty we arrive at the contradiction:

$$|\tilde{\lambda}| + |\tilde{\mu}| + |\tilde{\nu}| = (k+d)(n-k-d) > (k+d)(n-k-d).$$

Therefore no such subspace W can exist; hence there is no curve C intersecting $\Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet)$ and $\Omega_\nu(H_\bullet)$ for general flags. We conclude that $\langle \Omega_\lambda, \Omega_\mu, \Omega_\nu \rangle_d = 0$.

Now consider the case $d > n - k$. Note that since

$$\ell(\lambda) + \ell(\mu) \leq k,$$

it follows that

$$|\lambda| + |\mu| \leq k(n-k).$$

We can then upper bound $|\lambda| + |\mu| + |\nu|$ by

$$|\lambda| + |\mu| + |\nu| \leq 2k(n-k) < k(n-k) + dn.$$

Thus by Proposition 5.4 it follows that $\langle \Omega_\lambda, \Omega_\mu, \Omega_\nu \rangle_d = 0$.

□

In the proof we used Lemma 5.7 to deduce the following corollary

Corollary 5.10. *As a corollary to Lemma 5.7 it follows that if $|\tilde{\lambda}| + |\tilde{\mu}| + |\tilde{\nu}| = (k+d)(n-(k+d))$ and*

$$\langle \Omega_{\tilde{\lambda}}(F_\bullet), \Omega_{\tilde{\mu}}(G_\bullet), \Omega_{\tilde{\nu}}(H_\bullet) \rangle_0 = 0 \quad \text{in } \text{Gr}(k+d, n)$$

then

$$\langle \Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet), \Omega_\nu(H_\bullet) \rangle_d = 0 \quad \text{in } \text{Gr}(k, n).$$

Proof. The ideas for this proof can be found in the proof of Proposition 5.9; we restate them here. If we have $|\tilde{\lambda}| + |\tilde{\mu}| + |\tilde{\nu}| = (k+d)(n-(k+d)) = \dim(\text{Gr}(k+d, n))$ then for generic flags $F_\bullet, G_\bullet, H_\bullet$, the intersection $\Omega_{\tilde{\lambda}}(F_\bullet) \cap \Omega_{\tilde{\mu}}(G_\bullet) \cap \Omega_{\tilde{\nu}}(H_\bullet)$ is a finite number of points. The number of points in the intersection is counted precisely by the degree zero Gromov-Witten invariant $\langle \Omega_{\tilde{\lambda}}(F_\bullet), \Omega_{\tilde{\mu}}(G_\bullet), \Omega_{\tilde{\nu}}(H_\bullet) \rangle_0$. So if the invariant is zero, it follows that the intersection is empty. By Lemma 5.7 it then follows that there cannot be a curve of degree d intersecting $\Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet)$ and $\Omega_\nu(H_\bullet)$. Therefore we conclude $\langle \Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet), \Omega_\nu(H_\bullet) \rangle_d = 0$.

□

We include Corollary 5.10 as future reference and will apply it to the following proposition:

Proposition 5.11. *Let μ, ν, λ be partitions in a $k \times (n - k)$ rectangle. Then*

$$\langle \Omega_\mu, \Omega_\lambda, \Omega_\nu \rangle_d = 0$$

for $d > \alpha$ where $\alpha = \min(\ell(\nu), \ell(\mu), \ell(\lambda))$, provided that either

- 1. $k \leq n - \alpha$, or*
- 2. $k = n$.*

Proof. We will proceed by proving the contrapositive. That is, if

$$\langle \Omega_\mu, \Omega_\lambda, \Omega_\nu \rangle_d \neq 0,$$

then either $d \leq \alpha$ or $n - \alpha < k < n$. First consider the case $d \leq n - k$. Suppose $\langle \Omega_\mu, \Omega_\lambda, \Omega_\nu \rangle_d \neq 0$, then applying Corollary 5.10 the intersection $\Omega_{\tilde{\mu}}(F_\bullet) \cap \Omega_{\tilde{\lambda}}(G_\bullet) \cap \Omega_{\tilde{\nu}}(H_\bullet)$ in $\text{Gr}(k + d, n)$ for general flags is non-empty. Without loss of generality let μ be such that $\alpha = \ell(\mu)$. The upper bound of the number total boxes in the Young diagrams for $\tilde{\lambda}, \tilde{\mu}, \tilde{\nu}$ is given by

$$(12) \quad |\tilde{\lambda}| + |\tilde{\nu}| + |\tilde{\mu}| \leq (k + d)(n - (k + d)).$$

This upper bound follows since if $\Omega_{\tilde{\mu}}(F_\bullet) \cap \Omega_{\tilde{\lambda}}(G_\bullet) \cap \Omega_{\tilde{\nu}}(H_\bullet) \neq \emptyset$ then, since the flags are general, the sum of the codimension of $\Omega_{\tilde{\lambda}}(F_\bullet)$, $\Omega_{\tilde{\mu}}(G_\bullet)$, and $\Omega_{\tilde{\nu}}(H_\bullet)$ must be greater than or equal to the dimension of $\text{Gr}(k + d, n)$. We can provide a lower bound by using the definition of $\tilde{\lambda}$. Recall that $\tilde{\lambda}$ is obtained from λ by removing the d left most columns of λ . Therefore in going from λ to $\tilde{\lambda}$ we remove at most dk boxes in the Young diagram. Then we get the inequality

$$(13) \quad \begin{aligned} |\tilde{\lambda}| + |\tilde{\nu}| + |\tilde{\mu}| &\geq |\lambda| - kd + |\nu| - kd + |\mu| - d\alpha, \\ &= (k + d)(n - (k + d)) + d^2 - d\alpha. \end{aligned}$$

Using the upper bound in (12) we find $d(d - \alpha) \leq 0$; it follows that $d \leq \alpha$.

Now consider the case $d > n - k$. Suppose $\langle \Omega_\mu, \Omega_\lambda, \Omega_\nu \rangle_d \neq 0$, then by Proposition 5.4, we know $|\mu| + |\nu| + |\lambda| = k(n - k) + nd$. We can provide an upper bound for $|\mu| + |\nu| + |\lambda|$ by bounding two of the three terms by the maximum

number of boxes a diagram can have, namely $k(n - k)$; then since α is the length of one of μ, ν, λ , we can bound the total sum by

$$|\mu| + |\nu| + |\lambda| \leq 2k(n - k) + \alpha(n - k).$$

We then use the equation $|\mu| + |\nu| + |\lambda| = k(n - k) + nd$ to write down the inequalities

$$\begin{aligned} k(n - k) + n(n - k) &< k(n - k) + nd \leq 2k(n - k) + \alpha(n - k) \\ \implies (n - k)^2 &< \alpha(n - k) \\ \implies n - \alpha &< k < n. \end{aligned} \quad \square$$

5.4 Proof of the Quantum Pieri and Giambelli Formula

Before we prove the theorem we state a helpful lemma that follows directly from the classical Pieri formula.

Lemma 5.12. *Let λ, μ, p be partitions in a $k \times (n - k)$ rectangle such that $|\lambda| + |\mu| + p = k(n - k)$. Then*

$$\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_0 = \begin{cases} 1 & \text{if } \lambda_i + \mu_{k-i} \leq n - k \leq \lambda_{i-1} + \mu_{k-i} \\ 0 & \text{otherwise} \end{cases}$$

Proof. Let λ be a partition in a $k \times (n - k)$ rectangle. From (7), the classical Pieri formula Theorem 2.6 tells us $\langle \Omega_\lambda, \Omega_{\mu^\vee}, \Omega_p \rangle_0 = 1$ for $\lambda_i \leq \mu_i \leq \lambda_{i-1}$ and $|\mu| = p + |\lambda|$, and $\langle \Omega_\lambda, \Omega_{\mu^\vee}, \Omega_p \rangle_0 = 0$ otherwise. The equality $|\lambda| = p + |\mu|$ guarantees we have the right dimension:

$$|\lambda| + |\mu^\vee| + p = |\lambda| + k(n - k) - |\nu| + p = k(n - k).$$

From the inequality $\lambda_i \leq \mu_i \leq \lambda_{i-1}$ we deduce

$$\lambda_i \leq \mu_i^\vee \leq \lambda_{i-1} \Rightarrow \lambda_i + \mu_{k-i}^\vee \leq n - k \leq \lambda_{i-1} + \mu_{k-i}^\vee.$$

$\square$

We proceed with the proof of the quantum Pieri formula.

Theorem 5.13 (Quantum Pieri Formula, Bertram). *If λ is contained in an $k \times (n - k)$ rectangle and $p \leq n - k$ then:*

$$\Omega_p \star \Omega_\lambda = \sum_{\substack{\lambda_i \leq \mu_i \leq \lambda_{i-1} \\ |\mu| = p + |\lambda|}} \Omega_\mu + q \left(\sum_{\substack{0 \leq \lambda_i - 1 \leq \nu_i \leq \lambda_{i-1} - 1 \\ |\nu| = p + |\lambda| - n}} \Omega_\nu \right)$$

Where the first sum is over all partitions μ that can be obtained from λ by adding p boxes, adding no two in the same column. The second sum is over all partitions ν that can be obtained from $\lambda - 1$ (the partition obtained from λ by deleting one box from each row) by adding $p - n + |\lambda|$ boxes. Observe that the second sum is zero for $\ell(\lambda) < k$ since we require $\lambda_i > 0$.

Proof. The first sum is given by the classical Pieri formula, coming from the classical part of the potential. Our focus will thus be on the $d > 0$ terms. Let ν be a partition fitting in a $k \times (n - k)$ rectangle. The Gromov-Witten invariants in the product $\Omega_p \star \Omega_\lambda$ are of the form

$$(14) \quad \langle \Omega_\lambda, \Omega_{\nu^\vee}, \Omega_p \rangle_d.$$

Let $\mu = \nu^\vee$. First let us handle the edge case: either μ or λ is the empty partition. Then $\alpha = \min(\ell(\lambda), \ell(p), \ell(\mu)) = 0$, so the conditions of Proposition 5.11 are satisfied for $d \neq 0$; therefore

$$\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_d = 0 \text{ for all } d \neq 0.$$

It is easy to verify that this matches the Quantum Pieri formula when either μ , or ν is the empty partition. Given that $\ell(p) = 1$ we may proceed assuming $\alpha = 1$.

Given $\alpha = 1$, Proposition 5.11 then tells us $\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_d = 0$ for $d > 1$. Let us now consider the case $d = 1$. We would like to apply Lemma 5.7 and do a similar analysis as in the proof of Proposition 5.11. However, to apply Lemma 5.7 we need $d \leq n - k$; hence we would need $k \neq n$. We will omit this edge-case to simplify our discussion, but the quantum Pieri formula can be easily verified in this case. Suppose $\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_1 \neq 0$, then consider $\langle \Omega_{\tilde{\lambda}}, \Omega_{\tilde{\mu}}, \Omega_{\tilde{p}} \rangle_0$ as in Corollary 5.10. We use the inequality (13) discussed in the proof of Proposition 5.11 and set $d = 1$. The inequality reads

$$|\tilde{\lambda}| + |\tilde{\mu}| + |\tilde{p}| \geq (k + 1)(n - (k + 1)).$$

The upper bound (12) forces this to be an equality. Recall that since

$\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_1 \neq 0$, we know $|\lambda| + |\mu| + p = k(n - k) + n$; therefore

$$|\tilde{\lambda}| + |\tilde{\mu}| + |\tilde{p}| = (k + 1)(n - (k + 1)) = (|\lambda| - k) + (|\mu| - k) + (p - 1).$$

It follows that $\tilde{\lambda}, \tilde{\mu}$ are obtained from λ, μ respectively by deleting k boxes each; this is only possible if $\ell(\lambda) = \ell(\mu) = k$. We conclude that $\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_1 = 0$ for $\ell(\lambda), \ell(\mu) \neq k$, as can be verified in the quantum Pieri formula.

We may proceed by assuming that $\ell(\lambda) = \ell(\mu) = k$. Note that the quantum Pieri formula can be rewritten as a condition on the Gromov-Witten invariants $\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_1$. If $|\lambda| + |\mu| + p = k(n - k) + n$ then the Quantum Pieri formula reads:

$$\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_1 = \begin{cases} 1 & \text{if } \lambda_i + \mu_{k-i} - 1 \leq (n - k) \leq \lambda_{i-1} + \mu_{k-i} - 1 \\ 0 & \text{otherwise.} \end{cases}$$

By substituting $\lambda_i = \tilde{\lambda}_i + 1$, and $\mu_i = \tilde{\mu}_i + 1$, we find the following equivalent inequalities:

$$(15) \quad \begin{aligned} & \lambda_i + \mu_{k-i} - 1 \leq (n - k) \leq \lambda_{i-1} + \mu_{k-i} - 1 \\ \iff & \tilde{\lambda}_i + \tilde{\mu}_{k-i} \leq n - (k + 1) \leq \tilde{\lambda}_{i-1} + \tilde{\mu}_{k-i} . \end{aligned}$$

Then by Lemma 5.12, we can phrase the $d = 1$ part of the Giambelli formula as $\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_1 = \langle \Omega_{\tilde{\lambda}}, \Omega_{\tilde{\mu}}, \Omega_{\tilde{p}} \rangle_0$. We proceed to show this equality. Since $\langle \Omega_{\tilde{\lambda}}, \Omega_{\tilde{\mu}}, \Omega_{\tilde{p}} \rangle_0$ can at most take the values 0 or 1, we only have two cases to check.

Consider the case $\langle \Omega_{\tilde{\lambda}}, \Omega_{\tilde{\mu}}, \Omega_{\tilde{p}} \rangle_0 = 0$; thus for general flags $F_\bullet, G_\bullet, H_\bullet$, the intersection $\Omega_{\tilde{\lambda}}(F_\bullet) \cap \Omega_{\tilde{\mu}}(G_\bullet) \cap \Omega_{\tilde{p}}(H_\bullet)$ is empty. Therefore by Lemma 5.10 there is no degree one curve intersecting $\Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet)$, and $\Omega_p(H_\bullet)$; hence $\langle \Omega_\lambda, \Omega_\mu, \Omega_p \rangle_1 = 0$.

Now consider the case $\langle \Omega_{\tilde{\lambda}}, \Omega_{\tilde{\mu}}, \Omega_{\tilde{p}} \rangle_0 = 1$. By Remark 2.9 it follows that $W \in \Omega_{\tilde{\lambda}}(F_\bullet) \cap \Omega_{\tilde{\mu}}(G_\bullet) \cap \Omega_{\tilde{p}}(H_\bullet)$, where $\dim(W) = k + 1$, must lie in the intersection of the open Schubert cells $W \in \Omega_{\tilde{\lambda}}^\circ(F_\bullet) \cap \Omega_{\tilde{\mu}}^\circ(G_\bullet) \cap \Omega_{\tilde{p}}^\circ(H_\bullet)$. In particular, by setting $i = k$ in (1) we know that

$$\dim(W \cap F_{n-\tilde{\lambda}_k-1}) = \dim(W \cap G_{n-\tilde{\mu}_k-1}) = k .$$

Recall $\tilde{\lambda}_k = \lambda_k - 1$, and $\tilde{\mu}_k = \mu_k - 1$. It follows that

$$F_{n-\tilde{\lambda}_k-1} = F_{n-\lambda_k} \quad \text{and} \quad G_{n-\tilde{\mu}_k-1} = G_{n-\mu_k} \quad .$$

Let $V_1 = W \cap F_{n-\lambda_k}$ and $V_2 = W \cap G_{n-\mu_k}$. One can further verify that $V_1 \in \Omega_\lambda(F_\bullet)$ and $V_2 \in \Omega_\mu(G_\bullet)$. Then note that the intersection $\Omega_\lambda(F_\bullet) \cap \Omega_\mu(G_\bullet)$ is empty unless $\lambda = \mu^\vee$ due to dimension constraints. In particular, having $\langle \Omega_{\tilde{\lambda}}, \Omega_{\tilde{\mu}}, \Omega_{\tilde{p}} \rangle_0 \neq 0$ implies that $|\tilde{\lambda}| + |\tilde{\mu}| + \tilde{p} = (k+1)(n - (k+1))$, which in turn tell us

$$|\lambda| + |\mu| = k(n - k) + (n - p).$$

Thus if $F_\bullet, G_\bullet$ are generic, the intersection $\Omega_\lambda(F_\bullet) \cap \Omega_\mu(G_\bullet)$ is non-empty only for $n = p$ and $\lambda = \mu^\vee$. Since $p \leq n - k$ we can only have $n = p$ if $k = 0$; we leave it to the reader to verify the quantum Pieri formula for $k = 0$. We proceed with $\Omega_\lambda(F_\bullet) \cap \Omega_\mu(G_\bullet) = \emptyset$, therefore $V_1 \neq V_2$. Since $V_1, V_2 \subset W$, and $\dim(W) = k + 1$, then from dimension counting it follows that $V_1 \cap V_2 = k - 1$. Let $S = V_1 \cap V_2$. Then $\mathbb{P}(W/S)$, the space of k -dimensional subspaces L such that $S \subset L \subset W$, is a line passing through $\Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet)$. By Exercise 6.9 in [harris1992algaic], $\mathbb{P}(W/S)$ is the unique line passing through $V_1 \in \Omega_\lambda(F_\bullet)$, $V_2 \in \Omega_\mu(G_\bullet)$. We will see that it also passes through $\Omega_p(H_\bullet)$. Note that

$$\dim(W \cap H_{n-k-p+1}) = 1.$$

Then we can take $x \in W \cap H_{n-k-p+1}$. We have $S \subset S \oplus \mathbb{C}x \subset W$. It can be verified that $S \oplus \mathbb{C}x \in \Omega_p(H_\bullet)$. Thus with $\mathbb{P}(W/S)$ we can deduce

$$\langle \Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet), \Omega_p(H_\bullet) \rangle_1 \geq 1.$$

Now we must show that $\mathbb{P}(W/S)$ is in fact the unique such line passing through $\Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet), \Omega_p(H_\bullet)$. We will proceed by showing that any line that passes through those three Schubert varieties must contain V_1 and V_2 . By Lemma 5.7 and the case assumption $\langle \Omega_{\tilde{\lambda}}, \Omega_{\tilde{\mu}}, \Omega_{\tilde{p}} \rangle_0 = 1$, it follows that the span of any line C must be W . Let $V \in C$ be a point passing through $\Omega_\mu(F_\bullet)$. Then by (1),

$$\dim(V \cap F_{n-\lambda_k}) \geq k.$$

Since $\dim(V) = k$, it must be the case that $V \subset F_{n-\lambda_k}$. Furthermore, W is the span of V , so in fact $V \subset W \cap F_{n-\lambda_k}$. Note that $\dim(W \cap F_{n-\lambda_k}) = \dim(V) = k$, so we conclude $V = W \cap F_{n-\lambda_k} = V_1$. We may do a similar argument for $V \in C$

intersecting $\Omega_\mu(G_\bullet)$, and conclude $V = V_2$. It follows that $C = \mathbb{P}(W/S)$. As a last comment, note that since 3-pointed degree 1 maps $\mathbb{P}^1 \rightarrow \text{Gr}(k, n)$ don't have automorphisms, thus the number of lines is the same as the number of maps. By lemma 4.2 we conclude that $\langle \Omega_\lambda(F_\bullet), \Omega_\mu(G_\bullet), \Omega_p(H_\bullet) \rangle_1 = 1$. $\square$

Before we move on to the proof of the Giambelli Formula, we state a helpful lemma:

Lemma 5.14. *Let $1 \leq j \leq k$ and $0 \leq i_j \leq n - k$. Then*

$$\Omega_{i_1} \star \Omega_{i_2} \star \cdots \star \Omega_{i_k} = \Omega_{i_1} \cdot \Omega_{i_2} \cdot \cdots \cdot \Omega_{i_k}.$$

Proof. First note that by the remark at the end of Theorem 5.13

$$\Omega_i \star \Omega_\mu = \Omega_i \cdot \Omega_\mu$$

for $\ell(\mu) < k$. Furthermore, $\Omega_i \cdot \Omega_\mu$ will be a sum of Ω_ν , where each ν must satisfy

$$\ell(\nu) \leq \ell(\mu) + 1.$$

The above follows because no two boxes can be added to the same column. In particular, the number of rows in the Young diagram can increase at most by one. Then it follows by induction that $\Omega_{i_1} \star \cdots \star \Omega_{i_j}$ will be a sum of Ω_ν , where each ν must satisfy $\ell(\nu) < k$ for $j < k$. Lastly

$$\begin{aligned} \Omega_{i_1} \star (\Omega_{i_2} \star \cdots \star \Omega_{i_j}) &= \Omega_{i_1} \star (\Omega_{i_2} \cdot \cdots \cdot \Omega_{i_j}) \\ &= \Omega_{i_1} \star \left(\sum \Omega_\nu \right) \\ &= \sum (\Omega_{i_1} \cdot \Omega_\nu). \end{aligned}$$

$\square$

Theorem 5.15 (Quantum Giambelli Formula, Bertram). *If the Young diagram of the partition λ is contained in an $k \times (n - k)$ rectangle, then the Schubert class Ω_λ in $QH^*Gr(k, n)$ is given by $\Omega_\lambda = \det(\Omega_{\lambda_i + j - i})_{1 \leq i, j \leq k}$.*

Proof. Since the determinant is a sum of terms of the form $\Omega_{i_1} \star \cdots \star \Omega_{i_k}$ for $1 \leq i_j \leq n - k$, by Lemma 5.14 we have $\Omega_{i_1} \star \cdots \star \Omega_{i_k} = \Omega_{i_1} \cdot \cdots \cdot \Omega_{i_k}$. Therefore the statement follows from the classical Giambelli formula. $\square$

6 Appendix A. Moduli Theory

6.1 Moduli Problems

In this section we provide a brief introduction to some of the language we use from moduli theory. For a more comprehensive introduction to moduli theory, see [Hos23]. The idea of moduli theory is to find a space X whose points provide a natural parametrization of our objects of interest; we call this classification problem a moduli problem. We refer to X as the *moduli space* of these objects. This may seem silly at first since the collection of all objects we wish to parametrize would, trivially, parametrize the objects of interest. A useful moduli space will carry geometry; this geometry comes from families of continuously varying objects.

Example 6.1. $\mathbb{P}^n$ is the moduli space of one-dimensional quotients of $\mathbb{C}^{n+1}$. More generally, $\text{Gr}(k, V)$ is the moduli space of k -dimensional quotients of V .⁴

Our goal is to parametrize a collection of objects X , where the objects in X have a notion of equivalence $\sim$. To formally define a moduli problem we need:

1. For each scheme S , a family of objects in X over S . By this we simply mean a collection of objects that we can call a family over S .
2. A notion of equivalence of families over S , denoted by $\sim_S$.
3. The family over $\text{Spec}(\mathbb{C})$ should be the set of objects X ; and the equivalence $\sim_{\text{Spec}(\mathbb{C})}$ should be the equivalence $\sim$ on X .
4. For a morphism $S \rightarrow T$, we have a pullback map from families over T to families over S .
5. The definition of families, equivalence of families, and pullback maps, should define a functor

$$\mathcal{M} : \text{Sch}^{op} \rightarrow \text{Sets}, \quad \mathcal{M}(S) = \{\text{equivalence classes of families of } S\}.$$

We call $\mathcal{M}$ the moduli functor.

The motivation for encoding a moduli problem in a presheaf comes from the Yoneda lemma. The Yoneda lemma allows us to identify the category of schemes as a full subcategory of the category of presheafs. In other words, a scheme M is

⁴Alternatively we could use *subspaces* instead of quotients. There is an isomorphism $\text{Gr}(k, V) \simeq \text{Gr}(n - k, V)$. See Example 6.3 for why we use quotients instead of subspaces.

determined, up to unique isomorphism, by functor $\text{Hom}(-, M) : \text{Sch}^{op} \rightarrow \text{Sets}$. This equivalence tells us that a presheaf $\mathcal{F}$, which is naturally isomorphic to $\text{Hom}(-, M)$, can be identified with the scheme M . The Yoneda lemma leads to the notion of a *fine moduli* space.

6.2 Fine Moduli Space

Definition 6.2. Let $\mathcal{M} : \text{Sch}^{op} \rightarrow \text{Sets}$ be a moduli functor. We say a scheme M is a *fine moduli space* for $\mathcal{M}$ if there exists a natural isomorphism $\text{Hom}(-, M) \simeq \mathcal{M}$.

Example 6.3. Let us apply the recipe for a moduli problem to the moduli of one-dimensional quotients of a fixed vector space V . Our objects X are the collection of one dimensional subspaces of V . In this case our equivalence is simply equality of subspaces, $L_1 \sim L_2 \Leftrightarrow L_1 = L_2$.

Question 6.4. *What are families of one-dimensional quotients of V ?*

We want to think of families over S as a continuously varying assignment of a vector space to each point in S . A natural notion of families is thus a vector bundle; or in the algebraic language, a locally free sheaf over S . However, we also want to think of the fibers of the vector bundle as coming from a quotients of V . To do this, we think of V as a vector bundle over $\text{Spec}(\mathbb{C})$. Then we pullback $V \rightarrow \text{Spec}(\mathbb{C})$ along the structure map $S \rightarrow \text{Spec}(\mathbb{C})$ to get a vector bundle over S .

$$\begin{array}{ccc} \pi^*V & \longrightarrow & V \\ \downarrow & & \downarrow \\ S & \xrightarrow{\pi} & \text{Spec}(\mathbb{C}) \end{array}$$

The locally free sheaf corresponding to the bundle π^*V is denoted as $V \otimes_{\mathbb{C}} \mathcal{O}_S$. This is analogous to our usual change of coefficients formula in the context of modules. We are led to the following notion of families:

$$\mathcal{M}(S) = \text{families over } S = \{\text{locally free rank 1 quotients } V \otimes_{\mathbb{C}} \mathcal{O}_S \twoheadrightarrow \mathcal{L}\}.$$

As a sanity check we see that when $S = \text{Spec}(\mathbb{C})$ then a family over S is simply a one-dimensional quotient of V . To turn $\mathcal{M}$ into a functor we must provide maps $\mathcal{M}(T) \rightarrow \mathcal{M}(S)$ for every map $S \rightarrow T$. We can get the maps $\mathcal{M}(T) \rightarrow \mathcal{M}(S)$ by pulling-back the families along the map $f : S \rightarrow T$:

$$V \otimes_{\mathbb{C}} \mathcal{O}_T \twoheadrightarrow \mathcal{L} \quad \mapsto \quad f^*(V \otimes_{\mathbb{C}} \mathcal{O}_T) \simeq V \otimes_{\mathbb{C}} \mathcal{O}_S \twoheadrightarrow f^*(\mathcal{L}).$$

The definition of the above map exploits the fact that pullbacks preserve surjections; hence our choice to use quotients instead of inclusions.

Claim 6.5. *For $\dim(V) = n + 1$, then $\mathbb{P}^n$ is a fine moduli space for $\mathcal{M}$.*

We will not verify this claim here. For a proof see [Stacks, Tag 089T] . However, let us recall its consequence. The claim means $\mathbb{P}^n$ represents the functor $\mathcal{M}$: for every scheme S , there is a natural bijection

$$\mathrm{Hom}(S, \mathbb{P}^n) \cong \mathcal{M}(S).$$

Concretely, morphisms $S \rightarrow \mathbb{P}^n$ correspond exactly to rank-1 locally free quotients of $V \otimes_{\mathbb{C}} \mathcal{O}_S$, for $\dim(V) = n + 1$.

6.3 Coarse Moduli Space

A moduli functor $\mathcal{M}$ does not always admit a fine moduli space. We replace the notion of a fine moduli space with the weaker notion of *coarse moduli space*. Instead of demanding $\mathrm{Hom}(-, M) \simeq \mathcal{M}$, we ask for a universal natural transformation $\eta : \mathcal{M} \rightarrow \mathrm{Hom}(-, M)$ that is a bijection on $\mathrm{Spec}(\mathbb{C})$.

Definition 6.6. Let $\mathcal{M} : \mathrm{Sch}^{op} \rightarrow \mathrm{Sets}$ be a moduli functor. We say M is a coarse moduli space for $\mathcal{M}$ if there exists a natural transformation $\eta : \mathcal{M} \rightarrow \mathrm{Hom}(-, M)$ satisfying the following properties:

1. $\eta(\mathrm{Spec}(\mathbb{C})) \simeq \mathrm{Hom}(\mathrm{Spec}(\mathbb{C}), M)$.
2. For any scheme N and any natural transformation $\mu : \mathcal{M} \rightarrow \mathrm{Hom}(-, N)$ there exists a unique morphism $f : M \rightarrow N$ such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{\eta} & \mathrm{Hom}(-, M) \\ & \searrow \mu & \swarrow f_* \\ & \mathrm{Hom}(-, N) & \end{array} .$$

The universal property of η ensures that a coarse moduli space for $\mathcal{M}$ is unique up to unique isomorphism.

Both coarse and fine moduli spaces serve as good models for moduli spaces. The key distinction is that a fine moduli space M comes equipped with a *universal family* $\mathcal{U} \in \mathcal{M}(M)$, which corresponds to the identity map $\mathrm{id}_M \in \mathrm{Hom}(M, M)$. Tautologically, any map $f : S \rightarrow M$ can be obtained by pulling back the identity

map along f . By the Yoneda lemma any family in $\mathcal{M}(S)$ is the pullback of the universal family $\mathcal{U}$ along a unique map $f : S \rightarrow M$. We say a family is universal if it satisfies precisely the property above:

Definition 6.7. A *universal family* $\mathcal{U}$ is a family over M such that every family over S is isomorphic to $f^*\mathcal{U}$ for a unique $f : S \rightarrow M$.

In fact, the existence of a universal family implies M is a fine moduli space.

Example 6.8. Let X be a projective scheme. The moduli of genus 0 stable n -pointed curves admits a fine moduli space $\overline{M}_{0,n}$, whereas the moduli of genus 0 stable n -pointed maps to X only admits a coarse moduli space $\overline{M}_{0,n}(X, \beta)$.

7 Appendix B. Intersection Theory

In this section we will provide a brief overview of basic concepts in intersection theory. For a thorough treatment on the subject see [Ful84]. Let X be a scheme.

Definition 7.1. A k -cycle on X is a finite formal sum $\sum n_V [V]$ of irreducible closed subvarieties $V \subset X$, with $n_V \in \mathbb{Z}$.

A rational function on a subvariety of X determines a cycle on X . The Chow group $A_k(X)$ is defined as the abelian group generated by k -cycles modulo the k -cycles coming from rational functions.

Definition 7.2. Let X be a scheme, let $X_1, \dots, X_n$ be irreducible components of X , and let $\eta_1, \dots, \eta_n$ be the generic points of $X_1, \dots, X_n$ respectively. The geometric multiplicity m_i of X_i in X is defined to be the length of $\mathcal{O}_{X, \eta_i}$.

The fundamental cycle of X , denoted by $[X]$, is the cycle

$$[X] = \sum m_i [X_i].$$

The Chow groups can be thought of as algebro-geometric analogues of the homology groups in topology. There is a cohomology theory associated to the Chow groups that goes by the name of *operational Chow*, denoted by $A^*(X)$ see [Ful84]. $A^*(X)$ has a product analogous to the cup product in cohomology; we denote this product by $\cup$. There is pairing between Chow groups and operational Chow analogous to the cap product

$$\cap : A^l \times A_k \rightarrow A_{k-l}.$$

When X is smooth and proper we have the analogous Poincare duality statement $A^k(X) \simeq A_{n-k}(X)$. Poincare duality is given by the cap product with the fundamental class,

$$\cap[X] : A^k \xrightarrow{\simeq} A_{n-k}.$$

Consequently, we may canonically identify homology classes with classes in cohomology. Given a subvariety $\Gamma \subset X$, representing a class in $A_*(X)$, we may associate to it a class $\gamma \in A^*(X)$ where $[X] \cap \gamma = [\Gamma]$.

Remark 7.3. For homogeneous varieties $X = G/P$ over $\mathbb{C}$ (e.g. the Grassmannian) the Chow groups coincide with the topological groups $A_k(X) \simeq H_{2k}(X)$, and $A^k(X) \simeq H^{2k}(X)$. See Chapter 4 in [EH16].

7.1 Pushforwards and Pullbacks

For a proper morphism $f : X \rightarrow Y$ there is an induced map $f_* : A_*(X) \rightarrow A_*(Y)$; we call f_* pushforward. Explicitly, the pushforward $f_* : A_k(X) \rightarrow A_k(Y)$ can be defined on the class of a subvariety $V \subset X$ as

$$(16) \quad f_*[V] = \begin{cases} [k(V) : k(f(V))] \cdot [f(V)] & \text{if } \dim f(V) = \dim V \\ 0 & \text{if } \dim f(V) < \dim V, \end{cases}$$

where $k(V)$ is the field of rational functions on V , and $[k(V) : k(f(V))]$ is the degree of the extension $k((f(V))) \subset k(V)$. If X is proper then the structure map $\pi : X \rightarrow \text{Spec}(\mathbb{C})$ gives us the map

$$\pi_* : A_0(X) \rightarrow A_0(\text{Spec}(\mathbb{C})) \simeq \mathbb{Z}.$$

The pushforward along the structure map receives the special notation $\pi_* := \int_X$. We can extend $\int_X$ to a map on $A_*(X)$ by defining $\int_X[A] = 0$ if $[A] \in A_k(X)$ for $k > 0$. When X is smooth, Poincare duality allows us to define this integral map in operational Chow. In a slight abuse of notation we write

$$\int_X : A^*(X) \rightarrow \mathbb{Z}, \quad \gamma \mapsto \int_X \gamma \cap [X].$$

In addition, there exists a pullback on operational Chow $f^* : A^*(X) \rightarrow A^*(Y)$. A key property of pullbacks and pushforwards is the projection formula:

$$(17) \quad f_*(f^*\alpha \cap [A]) = \alpha \cap f_*[A].$$

We now discuss pullbacks on the Chow groups. If $f : X \rightarrow Y$ is a flat map, we have a pullback map $f^* : A_*(Y) \rightarrow A_*(X)$ defined on a subvariety $W \subset Y$ by $f^*[W] = [f^{-1}(W)]$. If X is smooth, we may alternatively pullback a class $[\Gamma] \in A_*(Y)$ by associating to it its corresponding class $\gamma \in A^*(Y)$, pullback that class to $A^*(X)$, and taking its corresponding Poincare dual in $A_*(X)$. When we run this procedure with a flat map we recover our original flat pullback $f^*[W] = [f^{-1}(W)]$. That is, given a flat map $f : X \rightarrow Y$ we have the following commutative diagram:

$$\begin{array}{ccc} A_*(Y) & \xrightarrow{f^*} & A_*(X) \\ \simeq \updownarrow & & \updownarrow \simeq \\ A^*(Y) & \xrightarrow{f^*} & A^*(X) \end{array} .$$

The top arrow is the flat pullback on Chow and the bottom arrow is the pullback on operational Chow.

7.2 Intersection Pairing

For this section we will assume X is smooth and projective. Our next task is to define an intersection product on $A_*(X)$. The intersection product will tell us how two subvarieties in “general position” intersect in X . We will define this intersection product using the *moving lemma*.

Lemma 7.4 (Moving lemma, [EH16]).

1. *Given classes $\alpha, \beta \in A_*(X)$ in the Chow group of a smooth, projective variety X , we can find representative cycles A and B intersecting transversely.*
2. *The class of the intersection of these cycles is independent of the choice of A and B .*

The preceding lemma allows us to define the intersection product of $\alpha \cdot \beta$ simply as the class of the scheme-theoretic intersection $A \cap B$. The second statement ensures that this operation is well-defined. It is important to note that this is not the approach taken by Fulton in [Ful84]. Indeed, according to [EH16], prior to the publication of Fulton’s book in 1984, there was no complete proof for the second part of the moving lemma. We have chosen this presentation to

keep the exposition simple and intuitive. The underlying idea is that the moving lemma captures the geometric intuition of the intersection product: it should fundamentally represent the intersection of varieties in “general position”. Thus, given subvarieties $\Gamma_1, \Gamma_2 \subset X$, we define $[\Gamma_1] \cdot [\Gamma_2] = [\Gamma'_1 \cap \Gamma'_2]$ where Γ'_1 and Γ'_2 are representatives of $[\Gamma_1]$ and $[\Gamma_2]$ respectively, such that their intersection is transverse. To see how the intersection product is defined for not necessarily smooth X see [Ful84]; the definition provided here agrees with that of [Ful84] in the case X is smooth. In Section 4, we assume that an intersection product exists on possibly non-smooth varieties. In particular, we assume the existence of the intersection product for $A_*(\overline{M}_{0,n}(X, \beta))$. Having defined the intersection product, we now define the intersection pairing $\langle [\Gamma_1], [\Gamma_2] \rangle$ as

$$\langle [\Gamma_1], [\Gamma_2] \rangle = \int_X [\Gamma_1] \cdot [\Gamma_2].$$

A key property of Poincare duality is that the cup product is Poincare dual to the intersection product. That is, if γ_1, γ_2 are the Poincare duals of $[\Gamma_1], [\Gamma_2]$, then

$$[\Gamma_1] \cdot [\Gamma_2] = (\gamma_1 \cup \gamma_2) \cap [X].$$

Therefore, we can define the intersection pairing on cohomology classes $\gamma_1, \gamma_2 \in A^*(X)$ as

$$\langle \gamma_1, \gamma_2 \rangle = \int_X \gamma_1 \cup \gamma_2.$$

When $[\Gamma_1] \in A_k(X)$ and $[\Gamma_2] \in A_{n-k}(X)$, the intersection pairing can be interpreted as counting the number of intersection points between the two subvarieties, taking multiplicity into account. These intersection pairings play an important role in our definition of quantum cohomology. As an early indication of their role in the quantum product, we will write the cup product between two basis elements of $A^*(X)$ using intersection pairings. Given a basis for $\gamma_1, \dots, \gamma_m \in A^*(X)$, we can write out the cup product $\gamma_i \cup \gamma_j$ as a linear combination of basis elements γ_e with coefficients $C_{i,j}^e$.

$$(18) \quad \gamma_i \cup \gamma_j = \sum_e C_{i,j}^e \gamma_e$$

Letting $g_{ij} = \int_X \gamma_i \cup \gamma_j$, we can cup both sides of (18) with γ_f and then hit both sides with $\int_X$ to get

$$(19) \quad \int_X \gamma_i \cup \gamma_j \cup \gamma_f = \sum_e C_{i,j}^e \int_X \gamma_e \cup \gamma_f = \sum_e C_{i,j}^e g_{ef}.$$

Let g^{ef} denote the entries of the matrix inverse of the $m \times m$ matrix g_{ef} . Applying the inverse g^{ef} to both sides we solve for the coefficients $C_{i,j}^e$:

$$C_{i,j}^e = \sum_f \int_X \gamma_i \cup \gamma_j \cup \gamma_f g^{ef}.$$

Finally, substituting back in (18) we conclude

$$(20) \quad \gamma_i \cup \gamma_j = \sum_{e,f} \left(\int_X \gamma_i \cup \gamma_j \cup \gamma_f \right) g^{ef} \gamma_e.$$

We see in Section 4 that the terms $\left(\int_X \gamma_i \cup \gamma_j \cup \gamma_f \right)$ correspond to degree 0 Gromov-Witten invariants; in the quantum product we allow for contributions of higher degree.

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